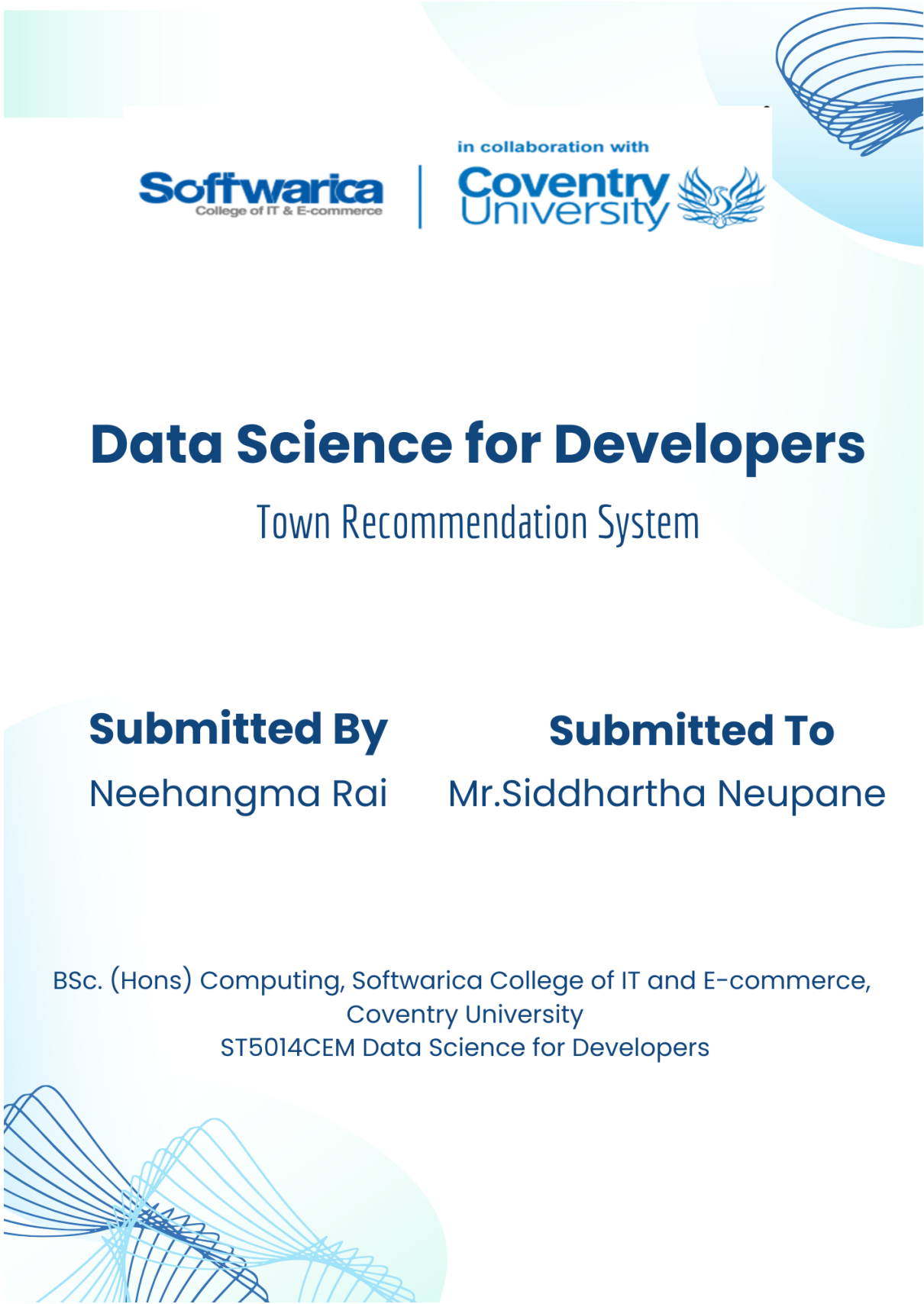




Softwarica
College of IT & E-commerce

in collaboration with

Coventry
University



Data Science for Developers

Town Recommendation System

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ST5014CEM Data Science for Developers

Table of Contents

Introduction	5
Data Cleaning.....	6
House Price Data Cleaning.....	7
Population and Postcode Data	8
Broadband Performance Data	9
Crime Data	10
Reason for Excluding the School Dataset.....	10
Exploratory Data Analysis.....	11
House Prices	12
Broadband Speed	16
Crime	21
Linear Model Visualization.....	27
House Price vs Download Speed for both District (Cheshire and Cumberland)	27
House prices vs drug offence rate 2023 both District (Cheshire and Cumberland) in a single diagram.....	29
Average download speed Vs. drug offence rate	31
Recommendation System.....	33
Overview	33
Result	33
Reflection	35
Overall Score	38
Legal and Ethical Issues	40
Conclusion	41
References.....	42
Appendix	43

Table Of Figures

Figure 1: Data Cleaning.....	6
Figure 2:House Price Dataset Cleaning code.....	7
Figure 3: Population and Post-Code Dataset Cleaning Code.....	8
Figure 4: Broadband Dataset Cleaning Code.....	9
Figure 5: Crime Dataset Cleaning Code	10
Figure 6: Exploratory Data Analysis	11
Figure 7: R Code of Line Graphs for Average House Prices (2022-2024) by District (Cheshire and Cumberland).....	12
Figure 8: Line Graphs for Average House Prices (2022-2024) by District Cheshire and Cumberland).....	13
Figure 9: R Code of Bar chart for Average House Prices (2022) by District (Cheshire and Cumberland).....	13
Figure 10: Bar chart for Average House Prices (2022) by District (Cheshire and Cumberland)	14
Figure 11: R Code for Box Plot for Average House Prices (2023) by District (Cheshire and Cumberland).....	14
Figure 12: Box plot for Average House Prices (2023) by District (Cheshire and Cumberland)	15
Figure 13: R Code of Boxplots of Broadband Speed Distribution by District(Cheshire).....	16
Figure 14: Boxplots for Broadband Speed Distribution by District(Cheshire)	17
Figure 15: R code of Boxplots for Broadband Speed Distribution by District (Cumberland)	17
Figure 16: Boxplots for Broadband Speed Distribution by District (Cumberland).....	18
Figure 17: R Code of Broadband Speed Distribution by District (Cheshire).....	18
Figure 18: Bar chart for Broadband Speed Distribution by District (Cheshire)	19
Figure 19: R Code of Bar chart for Broadband Speed Distribution by District (Cumberland)	19
Figure 20: Bar chart for Broadband Speed Distribution by District (Cumberland).....	20
Figure 21: R Code for Boxplot of Monthly Drug Offense Counts by District (Cheshire).....	21
Figure 22: Boxplot of Monthly Drug Offense Counts by District (Cheshire)	22
Figure 23: R Code for Boxplot of Monthly Drug Offense Counts by District (Cumberland)	22
Figure 24: Boxplot of Monthly Drug Offense Counts by District (Cumberland)	23
Figure 25: Code of Pie chart for Robbery rate for Cheshire	23
Figure 26: Pie chart for Robbery rate for Cheshire	24
Figure 27: Code of Radar chart for Vehicle crime rate for Cheshire	24
Figure 28: Radar chart for Vehicle crime rate for Cheshire	25
Figure 29: Code for Line chart for Drug offense rates per 10,000 people for both counties..	26
Figure 30: Line chart for Drug offense rates per 10,000 people for both counties.....	26
Figure 31: R Code of House Price vs Download Speed for both District (Cheshire and Cumberland).....	27

Figure 32: Graph of House Price vs Download Speed for both District (Cheshire and Cumberland).....	28
Figure 33: Linear model summary report and correlation of House Price Vs. Download Speed	28
Figure 34: R Code of house prices vs drug offence rate 2023 both District (Cheshire and Cumberland) in a single diagram	29
Figure 35: Graph of house prices vs drug offence rate 2023 both District (Cheshire and Cumberland) in a single diagram	30
Figure 36: Linear model summary report and correlation of House Price Vs. Drug Offence Rate.....	30
Figure 37: R Code of Average download speed Vs. drug offence rate	31
Figure 38: Graph of Average download speed Vs. drug offence rate	31
Figure 39: Linear model summary report and correlation of Average download speed Vs. Drug Offence rate	32
Figure 40: Top 10 House Score.....	33
Figure 41: Top 10 Crime Score	34
Figure 42: Top 10 Broadband Speed Score	34
Figure 43: House Price Recommendation.....	36
Figure 44: Broadband Speed score and Recommendation	37
Safety is measured on a scale from 0 to 10, with 10.00 being the safest. These areas have very low crime rates, which gives them a "Safe" rating. Figure 45: Crime (Safe Town) Recommendation.....	37
Figure 46: Overall Score.....	38

Introduction

The rapid increase in available data has made it easier to make informed decisions based on real evidence in areas such as real estate, infrastructure planning, and regional investment. For people and groups looking to find good places to buy property, depending only on guesswork or stories from others is not enough anymore. Instead, they need to use data in a smart way to compare different areas based on several important factors.

This project creates a system that uses data to help people decide where to invest in property in certain counties in England. The people who will use this system include investors, both from inside and outside the country, who need a clear and fair way to compare different areas based on real facts. Using data science is important here because there is a lot of data, and it comes from many different places.

The data varies in its measurement and requires cleaning before it can be used properly. By putting all these data sources together and looking at them closely, the project helps to give a full picture of how good a place is for investment.

The system looks at four main factors from the data:

- House prices, which show how much you can afford and how much the property is worth
- Crime rates, which show how safe a place is and how stable the community is
- Broadband speed, which shows how good the internet connection is
- Population distribution and postcode-to-LSOA mapping, Population distribution and the mapping of postcodes to LSOA areas were used to connect postcode-based data with standard LSOA codes and towns. This made it possible to consistently group data, perform analysis while keeping personal information private, and accurately combine population, crime, broadband, and housing data across Cheshire and Cumberland. ([*Find Open Data - data.gov.uk, n.d.*](#))

Each of these factors is adjusted so they can be compared fairly, and then they are added together to make an overall score. This score lets the system rank the different counties. The final result is a list of counties in order, along with a clear suggestion about which area is best for investment based on all the data studied.

Data Cleaning

Datasets collected from various public sources were not ready to use right away because they had issues like different formats, missing data, varying levels of detail, and extra, unnecessary information. To fix this, each dataset was cleaned and prepared separately before being combined into one consistent system for analysis. All the cleaning was done using the R programming language, mainly with tools from the tidy verse package group. ([Institute of Data, 2024](#))

The goal of cleaning the data was to make it better quality, easier to compare, and more reliable, while still keeping the important information needed for making investment decisions. ([Wikipedia contributors, 2025](#))

Figure 1: Data Cleaning

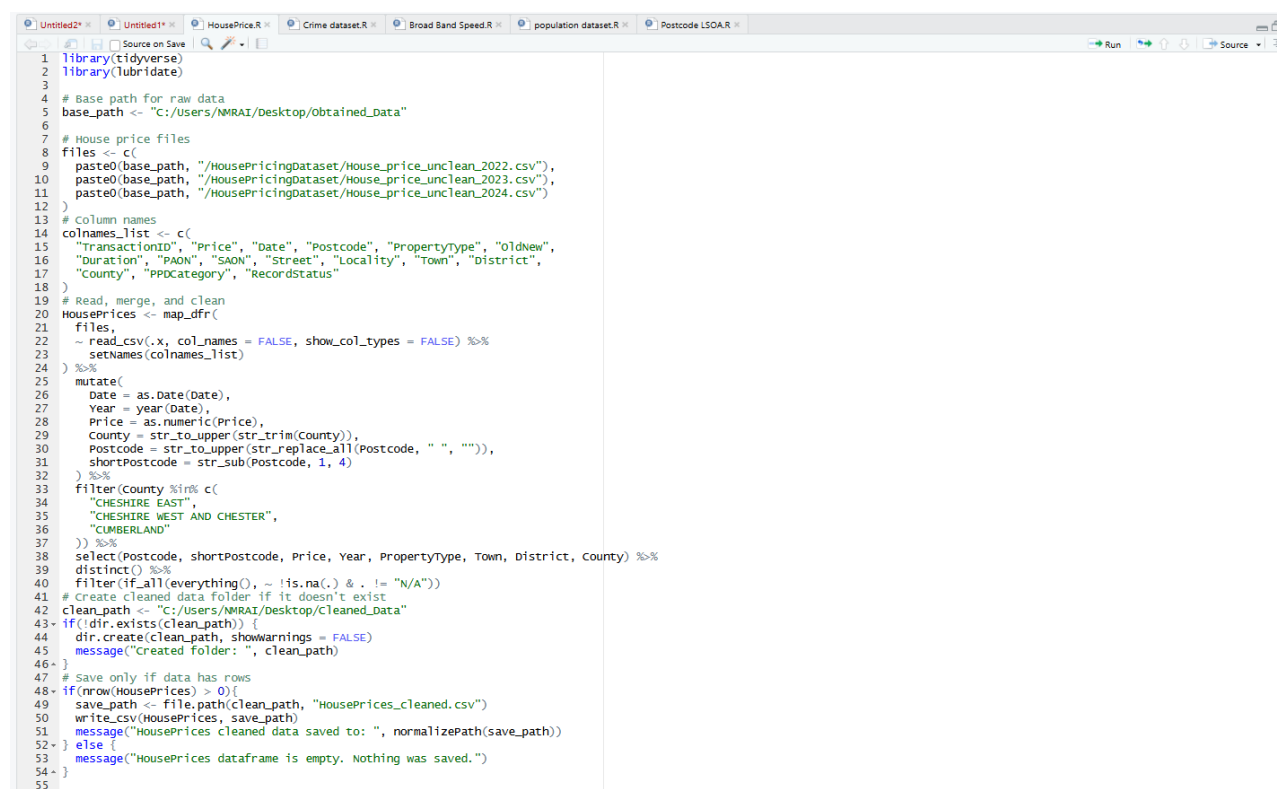


House Price Data Cleaning

The house price data used in this study was gathered from 2022 to 2024 and brought into R using CSV files. Before putting the files together, I checked that each dataset had the same structure so that variables would match up properly. Once everything was aligned, the datasets were combined into one table to make it easier to analyse data across different years. To keep the analysis focused on the study's goal, I only included data from Cheshire East, Cheshire West and Chester, and Cumberland. This helped remove data from other areas that weren't relevant and made the dataset less cluttered.

After that, several steps were taken to prepare the data. Dates of transactions were used to create a new variable for the year, which made it easier to compare data over time. County and district names were made the same across all datasets to prevent duplicates caused by different ways of writing them. Any records with missing or wrong house price information were taken out because they could make summaries and graphs misleading. Once this was done, the cleaned dataset was saved for future use.

Figure 2: House Price Dataset Cleaning code



```

1 library(tidyverse)
2 library(lubridate)
3
4 # Base path for raw data
5 base_path <- "C:/Users/NMRAI/desktop/obtained_data"
6
7 # House price files
8 files <- c(
9   paste0(base_path, "/HousePricingdataset/House_price_unclean_2022.csv"),
10  paste0(base_path, "/HousePricingdataset/House_price_unclean_2023.csv"),
11  paste0(base_path, "/HousePricingdataset/House_price_unclean_2024.csv")
12 )
13
14 # Column names
15 colnames_list <- c(
16   "TransactionID", "Price", "Date", "Postcode", "PropertyType", "oldnew",
17   "Duration", "PAON", "SAON", "Street", "Locality", "Town", "District",
18   "County", "PPDCategory", "RecordStatus"
19 )
20
21 # Read, merge, and clean
22 HousePrices <- map_dfr(
23   files,
24   ~ read_csv(., col_names = FALSE, show_col_types = FALSE) %>%
25     setNames(colnames_list)
26 ) %>%
27 mutate(
28   Date = as.Date(Date),
29   Year = year(Date),
30   Price = as.numeric(Price),
31   County = str_to_upper(str_trim(County)),
32   Postcode = str_to_upper(str_replace_all(Postcode, " ", "")),
33   shortPostcode = str_sub(Postcode, 1, 4)
34 ) %>%
35 filter(County %in% c(
36   "CHESHIRE EAST",
37   "CHESHIRE WEST AND CHESTER",
38   "CUMBERLAND"
39 )) %>%
40 select(Postcode, shortPostcode, Price, Year, PropertyType, Town, District, County) %>%
41 distinct() %>%
42 filter(if_all(everything(), ~ !is.na(.) & . != "N/A"))
43
44 # Create cleaned data folder if it doesn't exist
45 clean_path <- "C:/Users/NMRAI/desktop/Cleaned_data"
46 if(!dir.exists(clean_path)) {
47   dir.create(clean_path, showWarnings = FALSE)
48   message("Created folder: ", clean_path)
49 }
50
51 # Save only if data has rows
52 if(nrow(HousePrices) > 0){
53   save_path <- file.path(clean_path, "HousePrices_cleaned.csv")
54   write_csv(HousePrices, save_path)
55   message("HousePrices cleaned data saved to: ", normalizePath(save_path))
56 } else {
57   message("HousePrices dataframe is empty. Nothing was saved.")
58 }
59

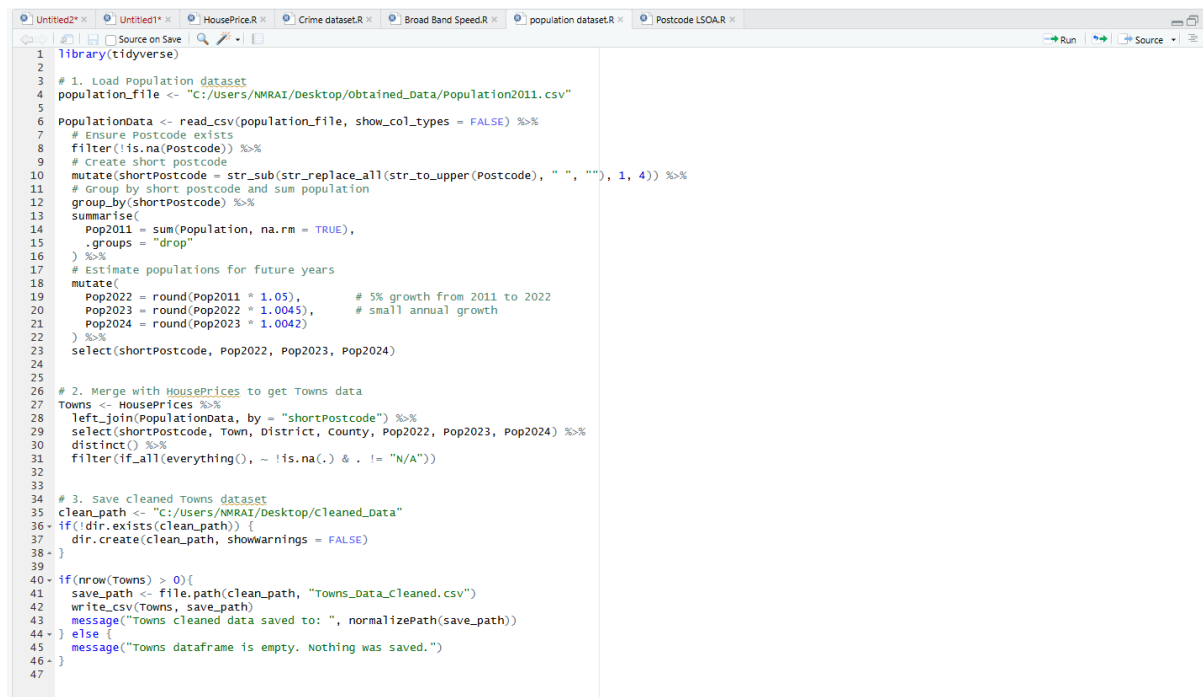
```

Population and Postcode Data

The population data gave information based on postcodes. Since full postcodes were too specific for comparing areas, I changed them to a shorter format to group areas more broadly. Population figures were then added up for each of these short postcodes to find the total number of people in each area.

To match up with the house price data, I used population estimates that covered the same time period, 2022 to 2024. This made it possible to combine population data with other datasets in a meaningful way. The cleaned population data was saved for later use.

Figure 3: Population and Post-Code Dataset Cleaning Code



```

1 library(tidyverse)
2
3 # 1. Load Population dataset
4 population_file <- "C:/Users/NMRAI/Desktop/Obtained_Data/Population2011.csv"
5
6 PopulationData <- read_csv(population_file, show_col_types = FALSE) %>%
7   # Ensure Postcode exists
8   filter(!is.na(Postcode)) %>%
9   # create short postcode
10  mutate(shortPostcode = str_sub(str_replace_all(str_to_upper(Postcode), " ", ""), 1, 4)) %>%
11  # Group by short postcode and sum population
12  group_by(shortPostcode) %>%
13  summarise(
14    Pop2011 = sum(Population, na.rm = TRUE),
15    .groups = "drop"
16  ) %>%
17  # Estimate populations for future years
18  mutate(
19    Pop2022 = round(Pop2011 * 1.05),      # 5% growth from 2011 to 2022
20    Pop2023 = round(Pop2022 * 1.0045),   # small annual growth
21    Pop2024 = round(Pop2023 * 1.0042)
22  ) %>%
23  select(shortPostcode, Pop2022, Pop2023, Pop2024)
24
25
26 # 2. Merge with HousePrices to get Towns data
27 Towns <- HousePrices %>%
28   left_join(PopulationData, by = "shortPostcode") %>%
29   select(shortPostcode, Town, District, County, Pop2022, Pop2023, Pop2024) %>%
30   distinct() %>%
31   filter(if_all(everything(), ~ !is.na(.)) & . != "N/A")
32
33
34 # 3. Save cleaned Towns dataset
35 clean_path <- "C:/Users/NMRAI/Desktop/Cleaned_Data"
36 if(!dir.exists(clean_path)) {
37   dir.create(clean_path, showWarnings = FALSE)
38 }
39
40 if(nrow(Towns) > 0){
41   save_path <- file.path(clean_path, "Towns_Data_Cleaned.csv")
42   write_csv(Towns, save_path)
43   message("Towns cleaned data saved to: ", normalizePath(save_path))
44 } else {
45   message("Towns dataframe is empty. Nothing was saved.")
46 }
47

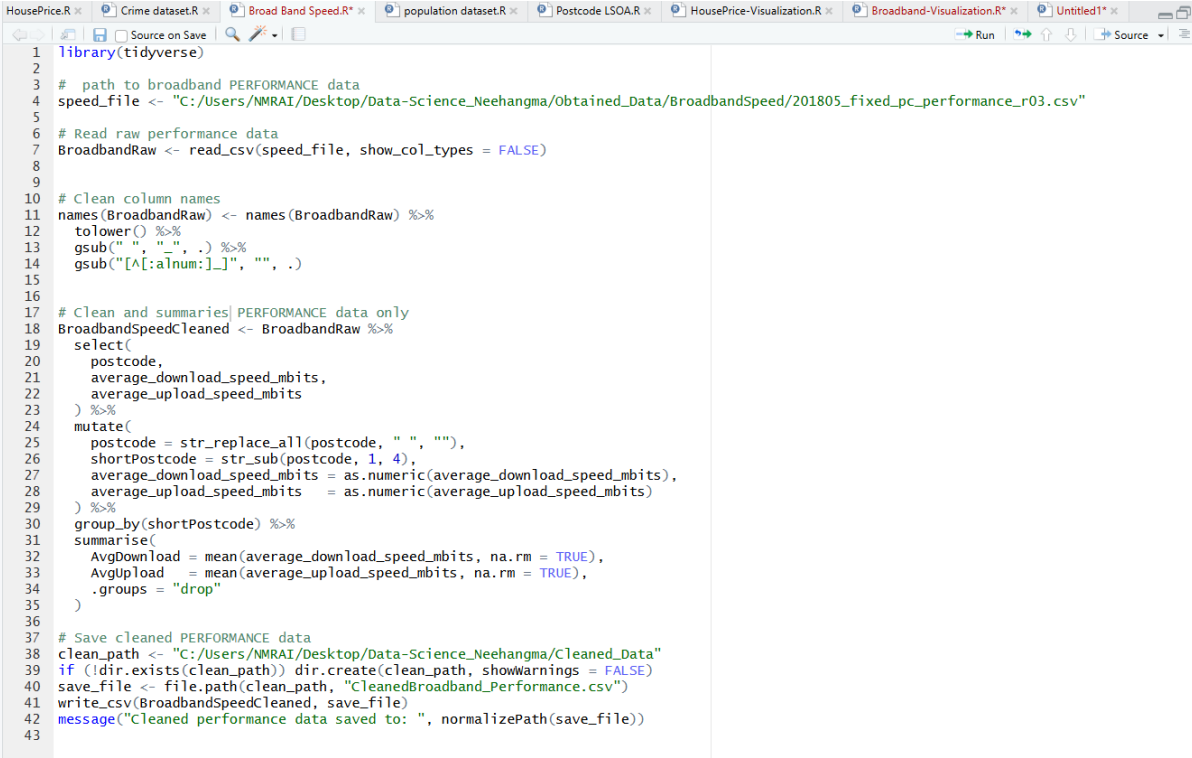
```

Broadband Performance Data

The broadband data was looked at to check the digital setup in the selected areas. When organizing the data, the names of the columns were made simpler and more standard so it was easier to understand. Only the information that really matters for how well the broadband works, like download speed, upload speed, and signs of fast internet, was kept. Any data that was missing or not relevant was taken out to keep the information accurate.

Then, the data was grouped by region so that we could compare how good the broadband is in each area. The cleaned broadband data was saved in a new file so it could be used for further checking and comparisons.

Figure 4: Broadband Dataset Cleaning Code



```

1 library(tidyverse)
2
3 # path to broadband PERFORMANCE data
4 speed_file <- "C:/Users/NMRAI/Desktop/Data-Science_Neehangma/Obtained_Data/BroadbandSpeed/201805_fixed_pc_performance_r03.csv"
5
6 # Read raw performance data
7 BroadbandRaw <- read_csv(speed_file, show_col_types = FALSE)
8
9
10 # Clean column names
11 names(BroadbandRaw) <- names(BroadbandRaw) %>%
12   tolower() %>%
13   gsub(" ", "_", .) %>%
14   gsub("[A-Za-z0-9:]", "", .)
15
16
17 # Clean and summaries PERFORMANCE data only
18 BroadbandSpeedCleaned <- BroadbandRaw %>%
19   select(
20     postcode,
21     average_download_speed_mbits,
22     average_upload_speed_mbits
23   ) %>%
24   mutate(
25     postcode = str_replace_all(postcode, " ", ""),
26     shortPostcode = str_sub(postcode, 1, 4),
27     average_download_speed_mbits = as.numeric(average_download_speed_mbits),
28     average_upload_speed_mbits = as.numeric(average_upload_speed_mbits)
29   ) %>%
30   group_by(shortPostcode) %>%
31   summarise(
32     AvgDownload = mean(average_download_speed_mbits, na.rm = TRUE),
33     AvgUpload = mean(average_upload_speed_mbits, na.rm = TRUE),
34     .groups = "drop"
35   )
36
37 # Save cleaned PERFORMANCE data
38 clean_path <- "C:/Users/NMRAI/Desktop/Data-Science_Neehangma/Cleaned_Data"
39 if (!dir.exists(clean_path)) dir.create(clean_path, showWarnings = FALSE)
40 save_file <- file.path(clean_path, "CleanedBroadbandPerformance.csv")
41 write_csv(BroadbandSpeedCleaned, save_file)
42 message("Cleaned performance data saved to: ", normalizePath(save_file))
43

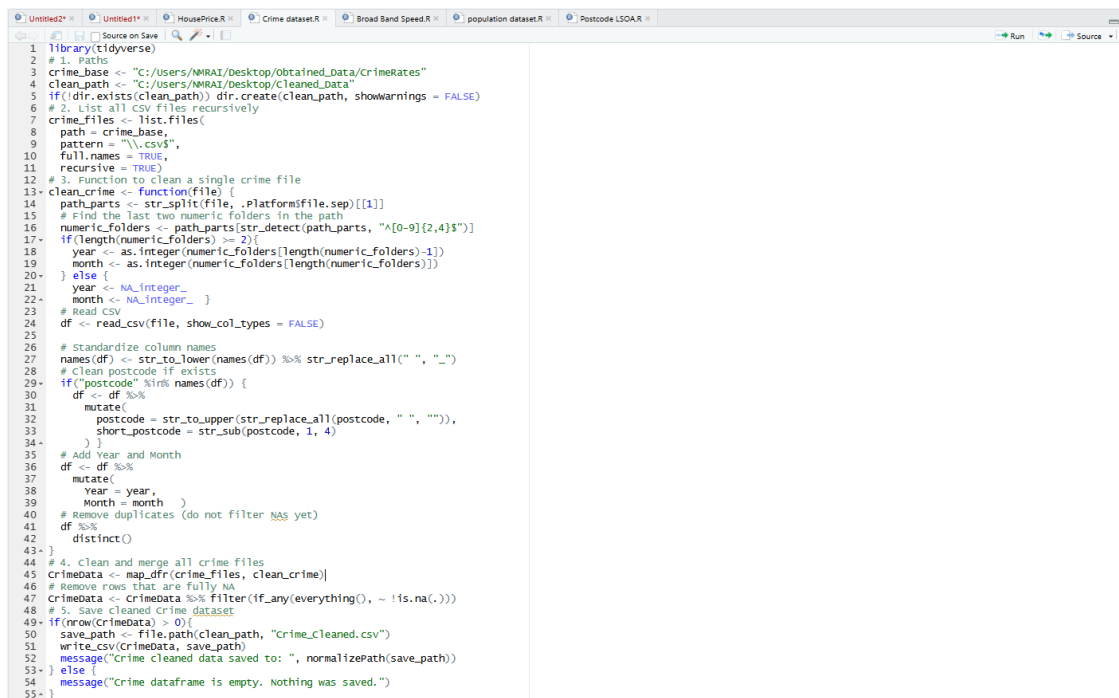
```

Crime Data

The crime data had detailed information about reported crimes. To get ready for analysis, only the parts needed to look at crime levels were kept, like where the crimes happened, what type of crime it was, and the results. To make the data cleaner, entries with missing info were removed, text was fixed so it was all the same, and repeats were taken out.

Then, the data was grouped by area to show clearer patterns of crime in each county. This cleaned crime data will be used later with population numbers to make fair comparisons between places.

Figure 5: Crime Dataset Cleaning Code



```

1 library(tidyverse)
2 # 1. Paths
3 crime_base <- "C:/Users/NMRAI/Desktop/Obtained_Data/CrimeRates"
4 clean_path <- "C:/Users/NMRAI/Desktop/Cleaned_Data"
5 if(!dir.exists(clean_path)) dir.create(clean_path, showWarnings = FALSE)
6 # 2. List all CSV files recursively
7 crime_files <- list.files(
8   path = crime_base,
9   pattern = "\\*.csv$",
10  full.names = TRUE,
11  recursive = TRUE)
12 # 3. Function to clean a single crime file
13 clean_crime <- function(file) {
14   path_parts <- str_split(file, .Platform$file.sep)[[1]]
15   # Find the last two numeric folders in the path
16   numeric_folders <- path_parts[str_detect(path_parts, "[0-9]{2,4}$")]
17   if(length(numeric_folders) >= 2){
18     year <- as.integer(numeric_folders[length(numeric_folders)-1])
19     month <- as.integer(numeric_folders[length(numeric_folders)])
20   } else {
21     year <- NA_integer_
22     month <- NA_integer_
23   }
24   # Read CSV
25   df <- read_csv(file, show_col_types = FALSE)
26   # Standardize column names
27   names(df) <- str_to_lower(names(df)) %>% str_replace_all(" ", "_")
28   # Clean postcode if exists
29   if("postcode" %in% names(df)) {
30     df <- df %>%
31       mutate(
32         postcode = str_to_upper(str_replace_all(postcode, " ", "")),
33         short_postcode = str_sub(postcode, 1, 4)
34       )
35   }
36   # Add Year and Month
37   df <- df %>%
38     mutate(
39       Year = year,
40       Month = month
41     )
42   # Remove duplicates (do not filter NAs yet)
43   df <- df %>%
44     distinct()
45 }
46 # 4. Clean and merge all crime files
47 crime_data <- map_dfr(crime_files, clean_crime)
48 # Remove rows that are fully NA
49 crime_data <- crime_data %>% filter(if_any(everything(), ~ !is.na(.)))
50 # 5. Save cleaned crime dataset
51 if(nrow(crime_data) > 0){
52   save_path <- file.path(clean_path, "Crime_Cleaned.csv")
53   write_csv(crime_data, save_path)
54   message("Crime cleaned data saved to: ", normalizePath(save_path))
55 } else {
56   message("Crime dataframe is empty. Nothing was saved.")
57 }

```

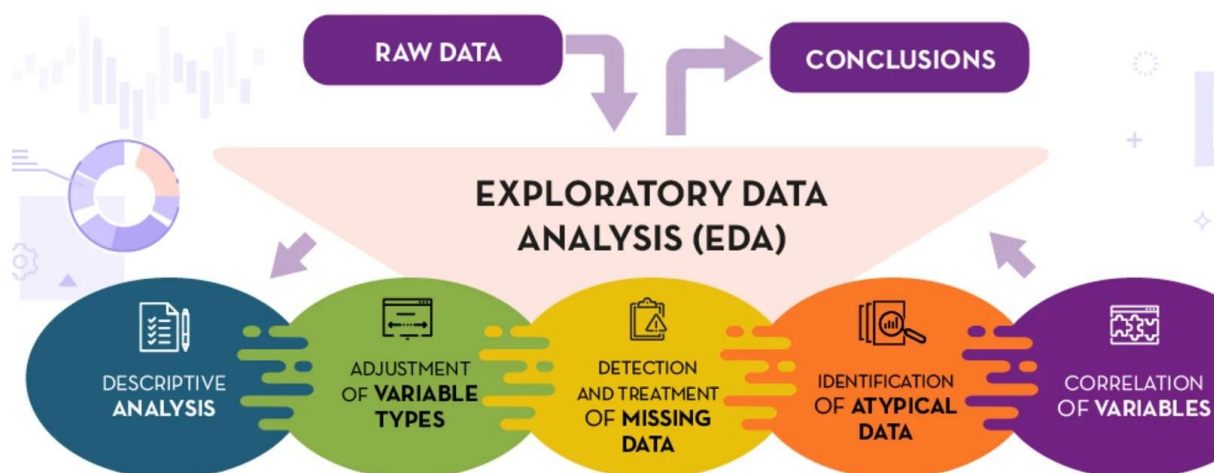
Reason for Excluding the School Dataset

The school dataset was left out of the analysis because there were problems with the data being consistent and easy to combine. Unlike the other datasets, it didn't have matching geographic details that could be accurately linked to towns or areas in Cheshire and Cumberland. Also, some important measures needed for a fair comparison were either missing or reported differently in different places. To prevent bias and keep the results of the exploratory analysis, linear modelling, and town rankings reliable, the study only used datasets that had a clear and consistent structure and coverage.

Exploratory Data Analysis

Exploratory Data Analysis was done to find out how house prices, broadband performance, and crime rates vary between different towns. Different types of charts were used, like boxplots to see how data is spread out and find any unusual numbers, bar charts to compare values between towns, pie charts to show the share of different crime types, radar charts to look at crime patterns from multiple angles, and line graphs to track how things change over time. These charts helped discover important information, check if the data was accurate, and plan for more detailed analysis and building of models. ([GeeksforGeeks, 2025b](#))

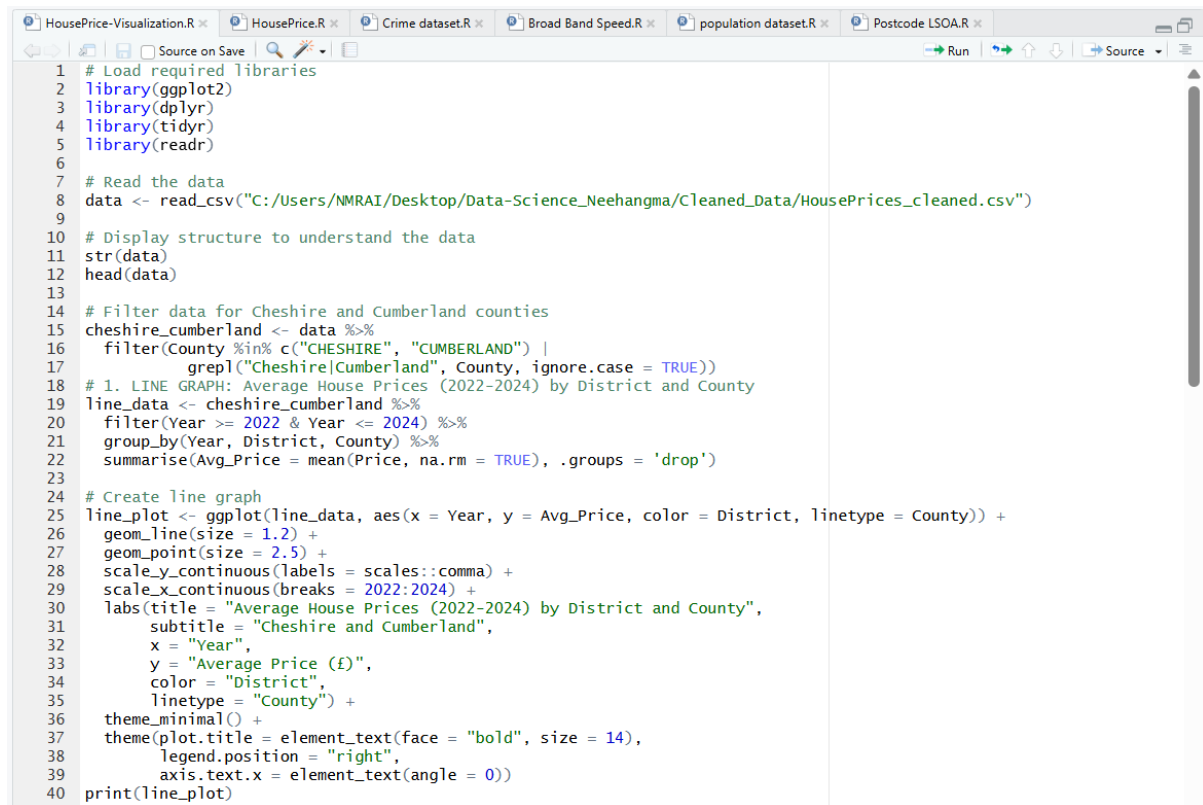
Figure 6: Exploratory Data Analysis



House Prices

The first look at house prices in Cheshire and Cumberland gave clear information about how property values are changing in these areas. Line graphs were used to show how average house prices changed from 2022 to 2024, helping to see how property values moved over time in both counties. Bar charts helped compare average house prices between different towns, and boxplots showed how house prices are spread out, how much they vary, and if there are any very high or very low prices. All together, these pictures gave a better idea of housing trends than just looking at average prices alone.

Figure 7: R Code of Line Graphs for Average House Prices (2022-2024) by District (Cheshire and Cumberland)



```

1 # Load required libraries
2 library(ggplot2)
3 library(dplyr)
4 library(tidyr)
5 library(readr)
6
7 # Read the data
8 data <- read_csv("C:/Users/NMRAI/Desktop/Data-Science_Neehangma/Cleaned_Data/HousePrices_cleaned.csv")
9
10 # Display structure to understand the data
11 str(data)
12 head(data)
13
14 # Filter data for Cheshire and Cumberland counties
15 cheshire_cumberland <- data %>%
16   filter(County %in% c("CHESHIRE", "CUMBERLAND") |
17         grepl("Cheshire|Cumberland", County, ignore.case = TRUE))
18 # 1. LINE GRAPH: Average House Prices (2022-2024) by District and County
19 line_data <- cheshire_cumberland %>%
20   filter(Year >= 2022 & Year <= 2024) %>%
21   group_by(Year, District, County) %>%
22   summarise(Avg_Price = mean(Price, na.rm = TRUE), .groups = 'drop')
23
24 # Create line graph
25 line_plot <- ggplot(line_data, aes(x = Year, y = Avg_Price, color = District, linetype = County)) +
26   geom_line(size = 1.2) +
27   geom_point(size = 2.5) +
28   scale_y_continuous(labels = scales::comma) +
29   scale_x_continuous(breaks = 2022:2024) +
30   labs(title = "Average House Prices (2022-2024) by District and County",
31        subtitle = "Cheshire and Cumberland",
32        x = "Year",
33        y = "Average Price (£)",
34        color = "District",
35        linetype = "County") +
36   theme_minimal() +
37   theme(plot.title = element_text(face = "bold", size = 14),
38         legend.position = "right",
39         axis.text.x = element_text(angle = 0))
40 print(line_plot)
41

```

Figure 8: Line Graphs for Average House Prices (2022-2024) by District Cheshire and Cumberland)

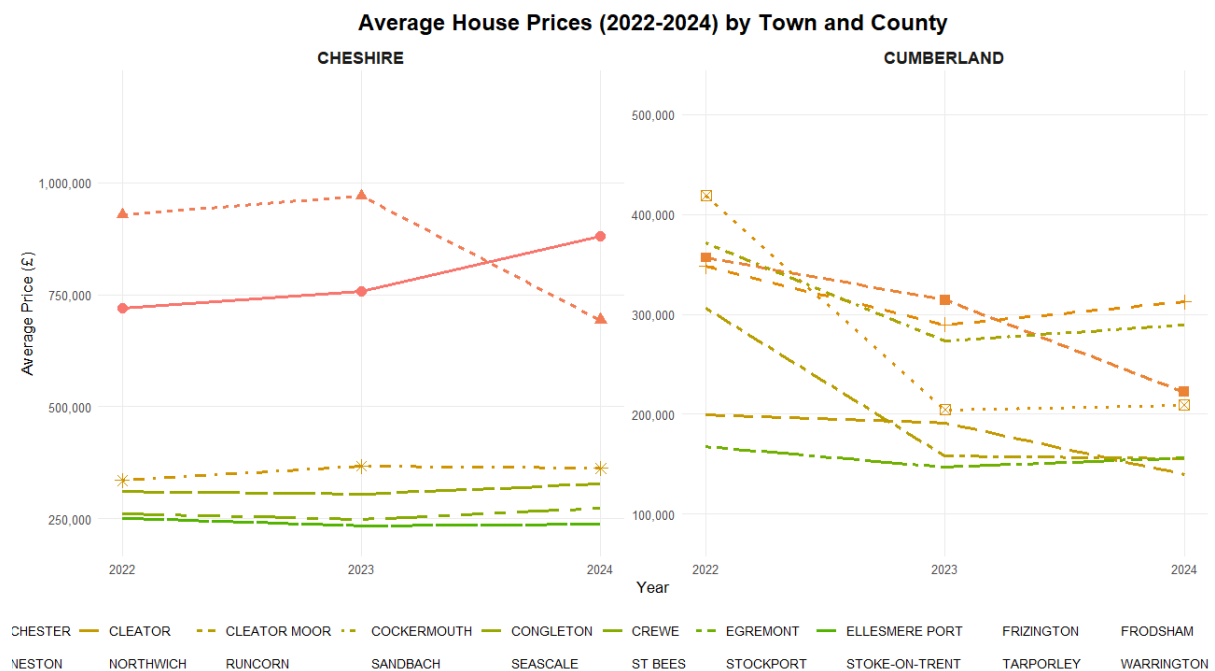


Figure 9: R Code of Bar chart for Average House Prices (2022) by District (Cheshire and Cumberland)

```
# -----
# BAR CHART: Average Prices in 2022
# -----

bar_data_processed <- data %>%
  filter(Year == 2022) %>%
  mutate(Parent_County = ifelse(grepl("Cheshire", County, ignore.case = TRUE),
                                "CHESHIRE", "CUMBERLAND")) %>%
  group_by(Parent_County, District, Town) %>%
  summarise(Avg_Price = mean(Price, na.rm = TRUE), .groups = 'drop') %>%
  group_by(Parent_County) %>%
  slice_max(order_by = Avg_Price, n = 8)

bar_chart <- ggplot(bar_data_processed, aes(x = reorder(Town, -Avg_Price), y = Avg_Price, fill = Town)) +
  geom_bar(stat = "identity", width = 0.7) +
  facet_wrap(~ Parent_County, scales = "free_x") +
  scale_y_continuous(labels = comma) +
  scale_fill_viridis_d(option = "viridis") +
  labs(title = "Average House Prices in 2022 by District and County",
       subtitle = "Cheshire and Cumberland",
       x = "Towns (Organized by District)",
       y = "Average Price (£)",
       fill = "Town Name") +
  theme_minimal() +
  theme(
    plot.title = element_text(face = "bold", size = 16, hjust = 0.5),
    strip.text = element_text(face = "bold", size = 12),
    axis.text.x = element_blank(),
    panel.grid.major.x = element_blank(),
    legend.position = "bottom",
    legend.title = element_text(face = "bold"),
    legend.text = element_text(size = 8)
  ) +
  guides(fill = guide_legend(nrow = 3, byrow = TRUE))

# Save Bar Chart
ggsave("C:/Users/NMRAI/Desktop/Data-Science_Neehangma/Graph/House Price-Graph/bar_chart_house_prices.png",
```

Figure 10: Bar chart for Average House Prices (2022) by District (Cheshire and Cumberland)

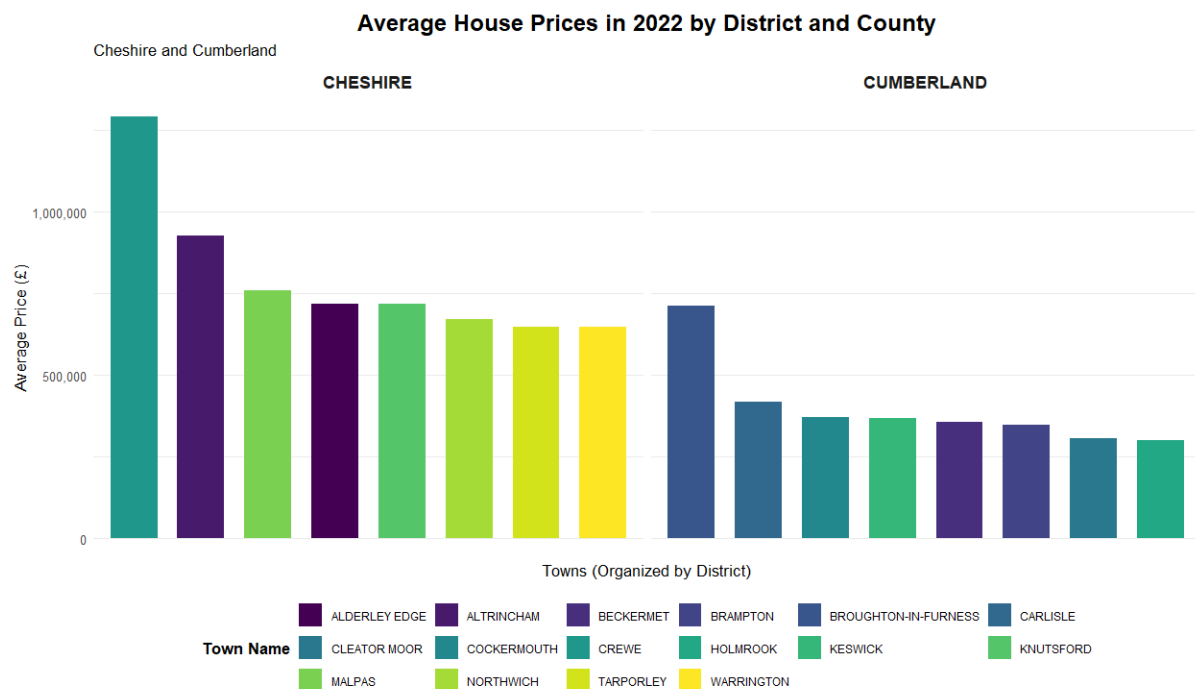


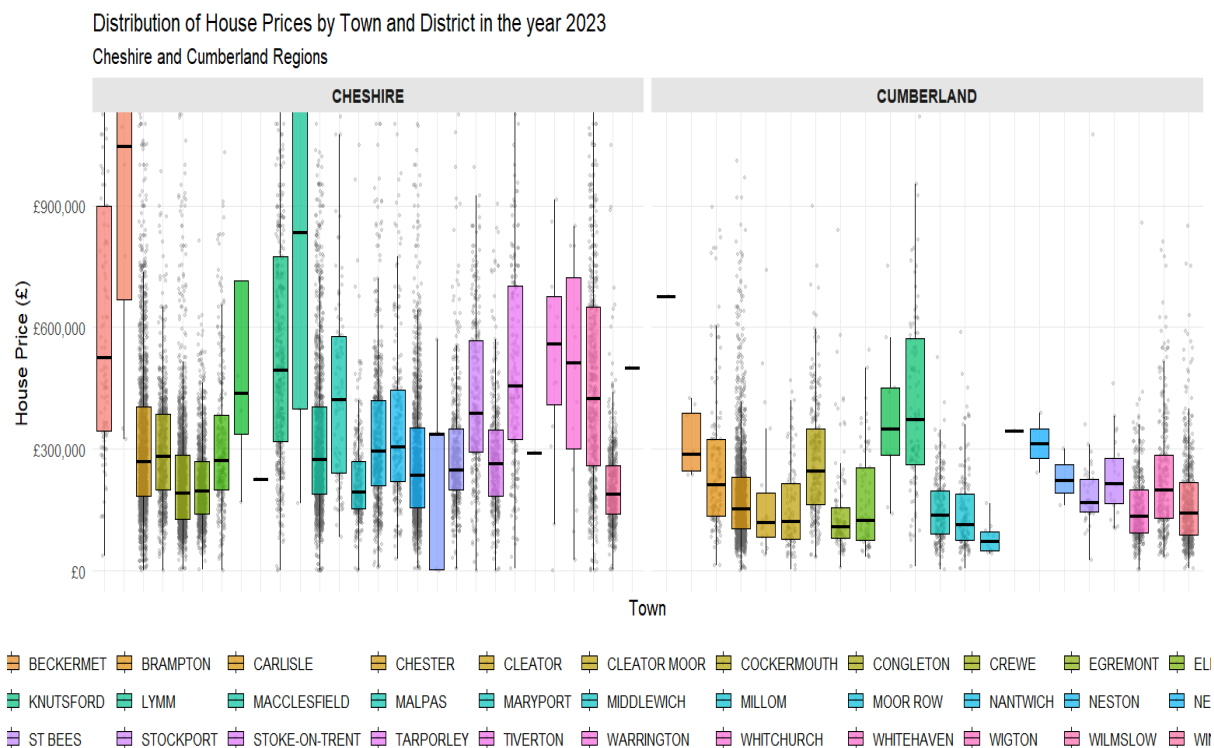
Figure 11: R Code for Box Plot for Average House Prices (2023) by District (Cheshire and Cumberland)

```
# BOX PLOT: Distribution in 2023
boxplot_data <- data %>%
  mutate(Parent_County = ifelse(grepl("Cheshire", County, ignore.case = TRUE),
                                "CHESHIRE", "CUMBERLAND")) %>%
  filter(Year == 2023)

box_plot <- ggplot(boxplot_data, aes(x = Town, y = Price, fill = Town)) +
  geom_jitter(width = 0.2, alpha = 0.2, size = 1, colour = "gray40") +
  geom_boxplot(alpha = 0.7, outlier.shape = NA, color = "black") +
  facet_wrap(~ Parent_County, scales = "free_x") +
  scale_y_continuous(labels = label_number(prefix = "£", big.mark = ",")) +
  coord_cartesian(ylim = c(0, quantile(boxplot_data$Price, 0.98, na.rm = TRUE))) +
  labs(title = "Distribution of House Prices by Town and District in 2023",
       subtitle = "Cheshire and Cumberland Regions",
       x = "Town",
       y = "House Price (£)") +
  theme_minimal(base_size = 13) +
  theme(
    strip.background = element_rect(fill = "gray90", color = NA),
    strip.text = element_text(face = "bold", size = 12),
    axis.text.x = element_blank(),
    legend.position = "bottom",
    legend.title = element_blank(),
    panel.grid.minor = element_blank()
  ) +
  guides(fill = guide_legend(nrow = 3, byrow = TRUE))

# Save Box Plot
ggsave("C:/Users/NMRAI/Desktop/Data-Science_Neehangma/Graph/House Price-Graph/boxplot_house_prices.png",
       plot = box_plot, width = 10, height = 7)
```

Figure 12: Box plot for Average House Prices (2023) by District (Cheshire and Cumberland)



Broadband Speed

The initial look at broadband speeds in Cheshire and Cumberland showed that internet performance varied between towns. Boxplots helped show how broadband speeds were spread out and how much they changed, pointing out places where speeds were always fast or slow. Stacked bar charts were also used to compare different speed groups across towns in both areas, making it easier to see how each speed level affected the overall availability of broadband. This helped give a better picture of the differences in digital infrastructure between the regions.

Figure 13: R Code of Boxplots of Broadband Speed Distribution by District (Cheshire)

```
# 2. Create boxplot with red outliers and diamond median
p_cheshire <- ggplot(cheshire_demo, aes(x = Town_Numbered, y = AvgSpeed, fill = Town)) +
  geom_boxplot(outlier.color = "red", outlier.shape = 16, outlier.size = 2, width = 0.6) +
  stat_summary(fun = median, geom = "point", shape = 23, size = 3.5,
    fill = "white", color = "black") +
  scale_fill_manual(values = c("#2471A3", "#5499C7", "#F39C12", "#16A085")) +
  labs(
    title = "Broadband Speed Distribution in CHESHIRE",
    x = "Town",
    y = "Average Download Speed (Mbps)"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(hjust = 0.5, face = "bold", size = 18, color = "#2C3E50"),
    axis.text.x = element_text(angle = 45, hjust = 1, face = "bold", size = 11),
    axis.title = element_text(face = "bold"),
    legend.position = "none",
    panel.grid.major.x = element_blank(),
    panel.grid.minor = element_blank(),
    axis.line = element_line(color = "black", linewidth = 0.8)
  )

# 3. Save and print
output_dir <- "C:/Users/NMRAI/Desktop/Data-Science_Neehangma/Graph/Broadband-Graph/"
if (!dir.exists(output_dir)) dir.create(output_dir, recursive = TRUE)

ggsave(file.path(output_dir, "Boxplot_Cheshire.png"), p_cheshire, width = 9, height = 7, dpi = 300)
print(p_cheshire)
```

Figure 14: Boxplots for Broadband Speed Distribution by District (Cheshire)

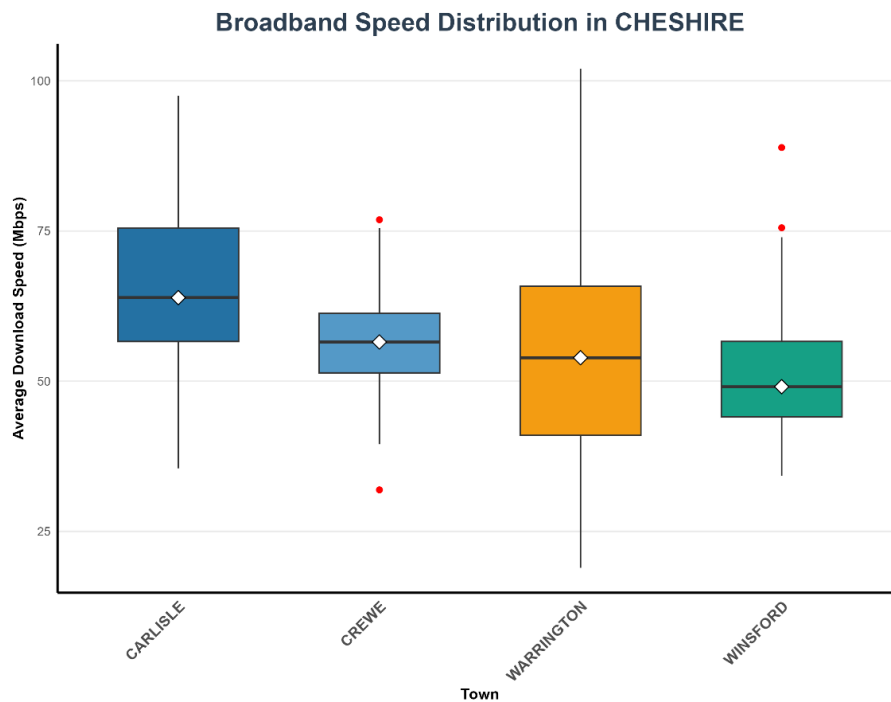


Figure 15: R code of Boxplots for Broadband Speed Distribution by District (Cumberland)

```
# 2. Create boxplot with red outliers and diamond median
p_cumberland <- ggplot(cumberland_demo, aes(x = Town_Numbered, y = AvgSpeed, fill = Town)) +
  geom_boxplot(outlier.color = "red", outlier.shape = 16, outlier.size = 2, width = 0.6) +
  stat_summary(fun = median, geom = "point", shape = 23, size = 3.5,
    fill = "white", color = "black") +
  scale_fill_manual(values = c("#C0392B", "#27AE60", "#F1C40F", "#7F8C8D")) +
  labs(
    title = "Broadband Speed Distribution in CUMBERLAND",
    x = "Town",
    y = "Average Download Speed (Mbps)"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(hjust = 0.5, face = "bold", size = 18, color = "#2C3E50"),
    axis.text.x = element_text(angle = 45, hjust = 1, face = "bold", size = 11),
    axis.title = element_text(face = "bold"),
    legend.position = "none",
    panel.grid.major.x = element_blank(),
    panel.grid.minor = element_blank(),
    axis.line = element_line(color = "black", linewidth = 0.8)
  )

# 3. Save and print
ggsave(file.path(output_dir, "Boxplot_Cumberland.png"), p_cumberland, width = 9, height = 7, dpi = 300)
print(p_cumberland)
```

Figure 16: Boxplots for Broadband Speed Distribution by District (Cumberland)

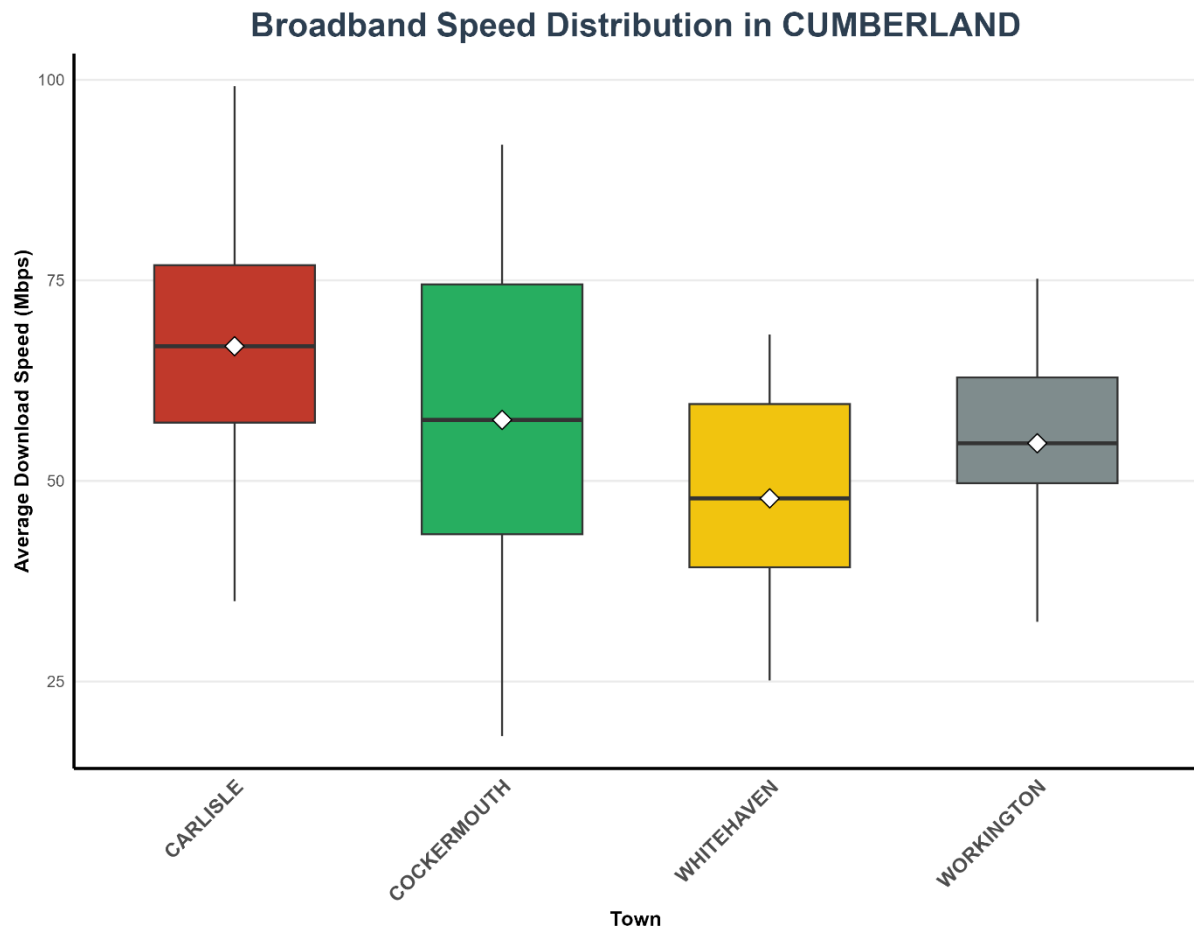


Figure 17: R Code of Broadband Speed Distribution by District (Cheshire)

```
# 3. Plot: CHESHIRE
p_cheshire_bar <- ggplot(cheshire_stacked, aes(x = Town_Numbered, y = Speed, fill = Type)) +
  geom_bar(stat = "identity") +
  scale_fill_manual(values = c("Base Speed" = "pink", "Additional Peak" = "gold")) +
  labs(
    title = "Broadband Speed Distribution in CHESHIRE",
    x = "Town",
    y = "Download Speed (Mbps)",
    fill = "Speed Component"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(hjust = 0.5, face = "bold", size = 16),
    axis.text.x = element_text(angle = 45, hjust = 1, face = "bold"),
    legend.position = "top"
  )
```

Figure 18: Bar chart for Broadband Speed Distribution by District (Cheshire)

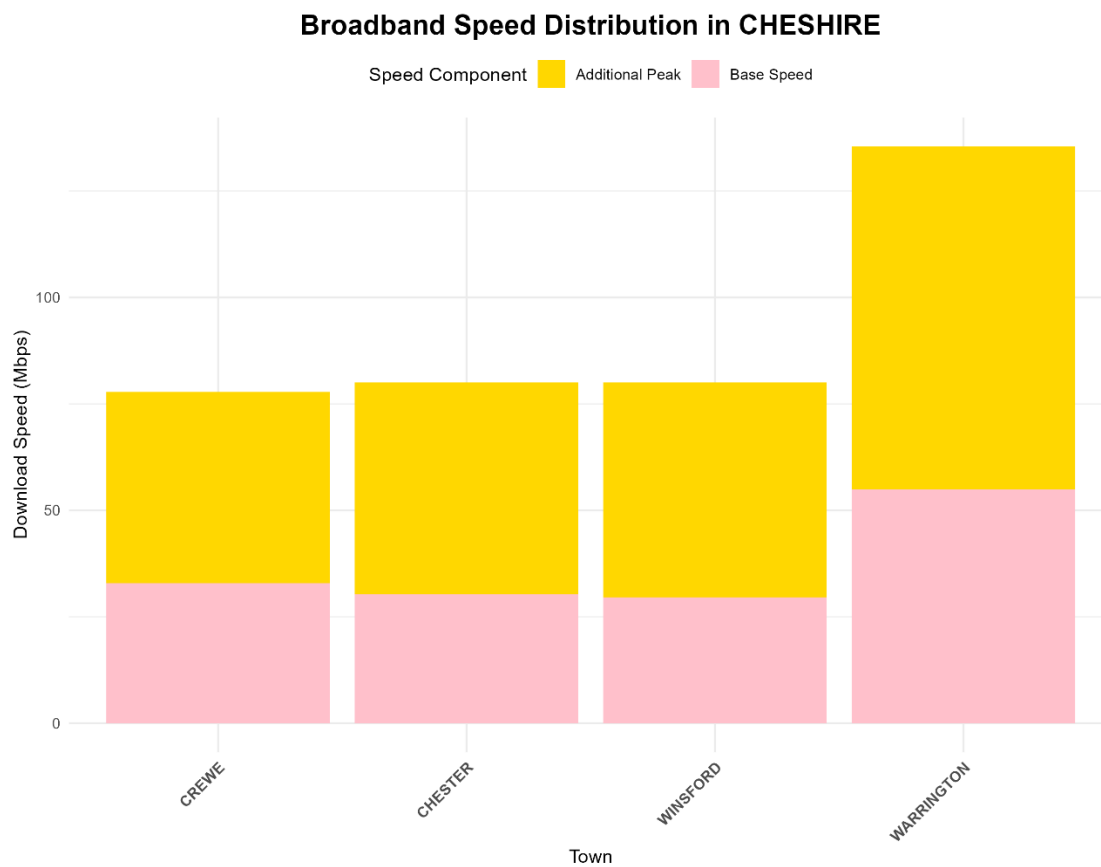


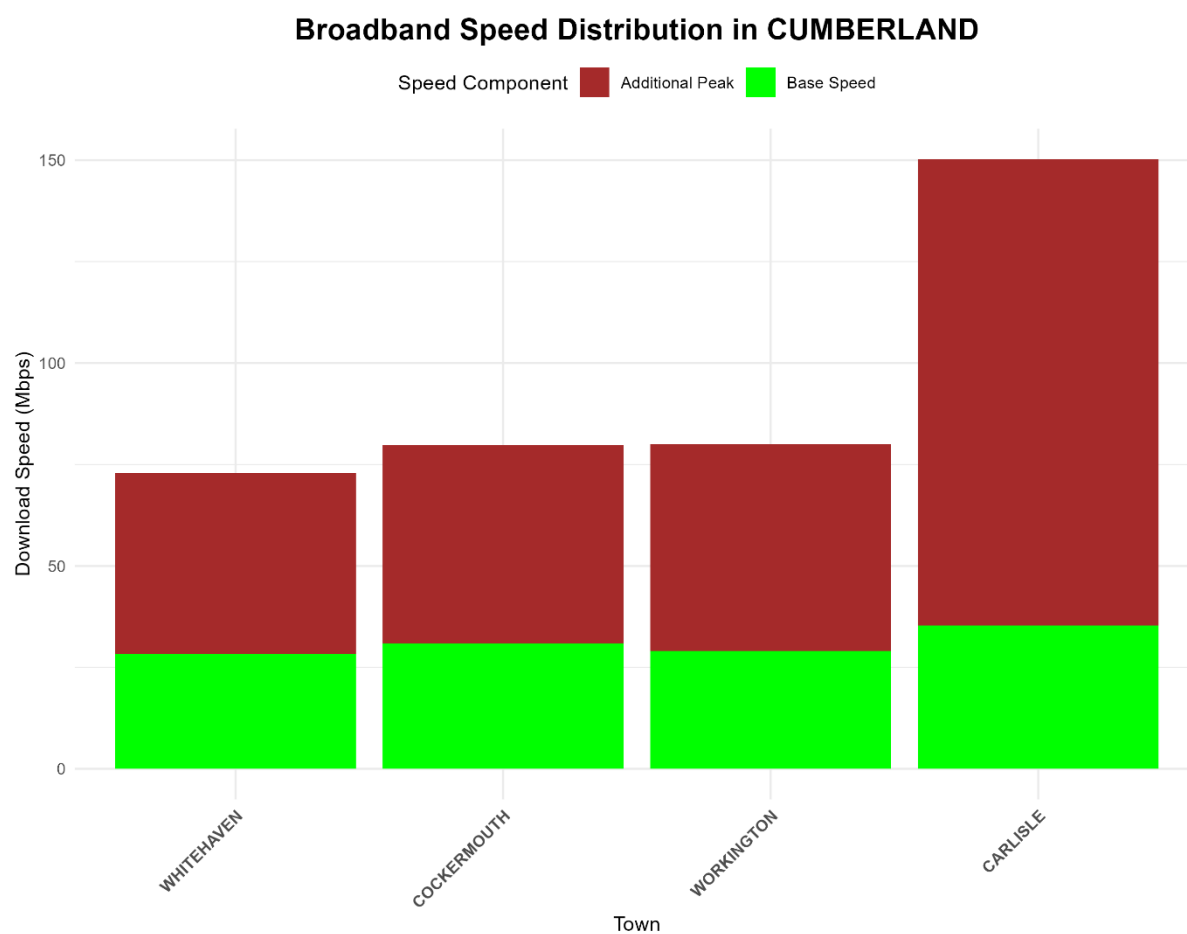
Figure 19: R Code of Bar chart for Broadband Speed Distribution by District (Cumberland)

```

# 4. Plot: CUMBERLAND
p_cumberland_bar <- ggplot(cumberland_stacked, aes(x = Town_Numbered, y = Speed, fill = Type)) +
  geom_bar(stat = "identity") +
  scale_fill_manual(values = c("Base Speed" = "green", "Additional Peak" = "brown")) +
  labs(
    title = "Broadband Speed Distribution in CUMBERLAND",
    x = "Town",
    y = "Download Speed (Mbps)",
    fill = "Speed Component"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(hjust = 0.5, face = "bold", size = 16),
    axis.text.x = element_text(angle = 45, hjust = 1, face = "bold"),
    legend.position = "top"
  )

```

Figure 20: Bar chart for Broadband Speed Distribution by District (Cumberland)



Crime

The study looked at crime trends in Cheshire and Cumberland, especially focusing on drug-related, vehicle-related, and robbery crimes. Boxplots were made to show how drug crime rates varied between towns and districts in each area, helping to spot differences and unusual cases. A radar chart was used to compare vehicle crime rates in 2023 across different locations for a specific month, making it easier to see how crimes compared in different ways. A pie chart showed the number of robberies per 10,000 people in 2023, giving a clear picture of how robbery cases were spread out and helping to understand the overall crime makeup. These charts made it easier to compare the two counties and helped pick out important factors for more detailed research later.

[\(GeeksforGeeks, 2025a\)](#)

Figure 21: R Code for Boxplot of Monthly Drug Offense Counts by District (Cheshire)

```
# BOX PLOT OF dDRUG OFFENCE RATE IN CHESHIRE AND CUMBERLAND
cat("\n=== CREATING DRUG BOX PLOTS ===\n")
plot_drug_boxplot <- function(county_name) {
  drugs <- crime_clean %>%
    filter(County == county_name,
           str_detect(crime_type, regex("Drug", ignore_case = TRUE))) %>%
    group_by(District, Date) %>%
    summarise(DrugCount = n(), .groups = "drop")
  cat(paste0("\n", county_name, " drug crimes found: ", nrow(drugs), " rows\n"))
  if(nrow(drugs) == 0) return(NULL)
  districts_with_data <- drugs %>%
    group_by(District) %>%
    summarise(TotalRecords = n(), MaxCount = max(DrugCount), .groups = "drop") %>%
    filter(MaxCount > 0) %>%
    pull(District)

  drugs_filtered <- drugs %>% filter(District %in% districts_with_data)
  top_districts <- drugs_filtered %>%
    group_by(District) %>%
    summarise(MedianCount = median(DrugCount), .groups = "drop") %>%
    arrange(desc(MedianCount)) %>%
    head(10) %>%
    pull(District)

  drugs_final <- drugs_filtered %>% filter(District %in% top_districts)
  p <- ggplot(drugs_final, aes(x = fct_reorder(District, DrugCount, median),
                              y = DrugCount, fill = District)) +
    geom_boxplot(alpha = 0.7, outlier.shape = 16, outlier.color = "red") +
    theme_minimal() +
    labs(
      title = paste(county_name, ": Monthly Drug Offense Count by District"),
      x = "District Name",
      y = "Drug Offense Count per Month"
    ) +
    theme(
      plot.title = element_text(hjust = 0.5, face = "bold"),
      axis.text.x = element_text(angle = 45, vjust = 1, hjust = 1),
      legend.position = "none"
    )
  print(p)
  plot_drug_boxplot("Cheshire")
  plot_drug_boxplot("Cumberland")
}
```

Figure 22: Boxplot of Monthly Drug Offense Counts by District (Cheshire)

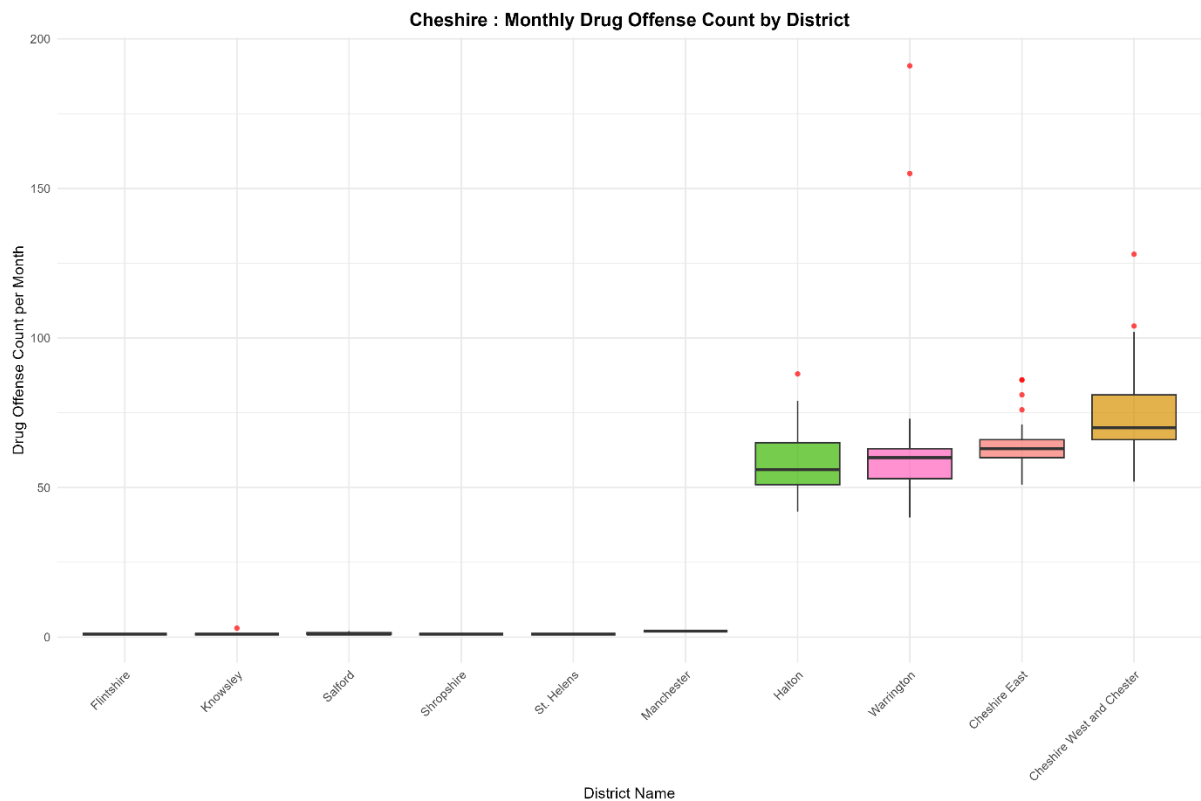


Figure 23: R Code for Boxplot of Monthly Drug Offense Counts by District (Cumberland)

```
# BOX PLOT OF dDRUG OFFENCE RATE IN CHESHIRE AND CUMBERLAND
cat("\n=== CREATING DRUG BOX PLOTS ===\n")
plot_drug_boxplot <- function(county_name) {
  drugs <- crime_clean %>%
    filter(County == county_name,
           str_detect(crime_type, regex("Drug", ignore_case = TRUE))) %>%
    group_by(District, Date) %>%
    summarise(DrugCount = n(), .groups = "drop")
  cat(paste0("\n", county_name, " drug crimes found: ", nrow(drugs), " rows\n"))
  if(nrow(drugs) == 0) return(NULL)
  districts_with_data <- drugs %>%
    group_by(District) %>%
    summarise(TotalRecords = n(), MaxCount = max(DrugCount), .groups = "drop") %>%
    filter(MaxCount > 0) %>%
    pull(District)

  drugs_filtered <- drugs %>% filter(District %in% districts_with_data)
  top_districts <- drugs_filtered %>%
    group_by(District) %>%
    summarise(MedianCount = median(DrugCount), .groups = "drop") %>%
    arrange(desc(MedianCount)) %>%
    head(10) %>%
    pull(District)

  drugs_final <- drugs_filtered %>% filter(District %in% top_districts)
  p <- ggplot(drugs_final, aes(x = fct_reorder(District, DrugCount, median),
                                y = DrugCount, fill = District)) +
    geom_boxplot(alpha = 0.7, outlier.shape = 16, outlier.color = "red") +
    theme_minimal() +
    labs(
      title = paste(county_name, ": Monthly Drug Offense Count by District"),
      x = "District Name",
      y = "Drug Offense Count per Month"
    ) +
    theme(
      plot.title = element_text(hjust = 0.5, face = "bold"),
      axis.text.x = element_text(angle = 45, vjust = 1, hjust = 1),
      legend.position = "none"
    )
  print(p)
  plot_drug_boxplot("Cheshire")
  plot_drug_boxplot("Cumberland")
}
```

Figure 24: Boxplot of Monthly Drug Offense Counts by District (Cumberland)

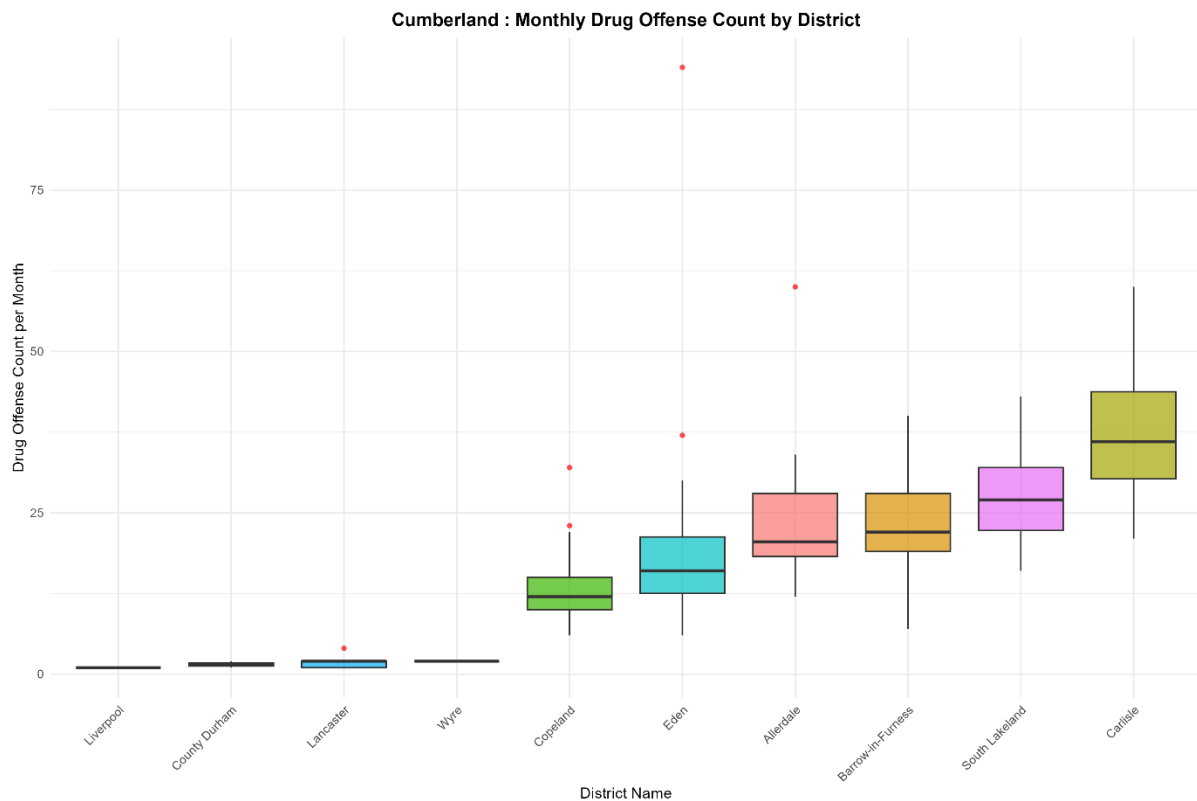


Figure 25: Code of Pie chart for Robbery rate for Cheshire

```
# Create pie chart
png(output_path, width = 1200, height = 800, res = 120, bg = "white")

par(mar = c(2, 2, 4, 2))

# Define distinct colors - one for each unique town
colors <- c("#FF6B6B", "#4ECDC4", "blue", "#FFA07A", "#98D8C8",
            "#F7DC6F", "#BB8FCE", "#85C1E2", "#F8B88B", "#AED6F1",
            "#A9DFBF", "#FAD7A0", "#D7BDE2", "#A3E4D7", "#F9E79F",
            "#FADB88", "#D5F4E6")

# Take only as many colors as needed
colors <- colors[1:nrow(robbery_rate_data)]

# Create labels with only percentages (to be shown on pie)
pie_labels <- paste0(round(robbery_rate_data$percentage, 1), "%")

# Create the pie chart
pie(robbery_rate_data$robbery_rate,
    labels = pie_labels,
    col = colors,
    border = "white",
    lwd = 2,
    cex = 1.2,
    font = 2)

# Add title
title(main = "Robbery Rate by Town (Cheshire, 2022-2024)",
      cex.main = 1.8,
      font.main = 2,
      line = 1)
```

Figure 26: Pie chart for Robbery rate for Cheshire

Robbery Rate by Town (Cheshire, 2022-2024)

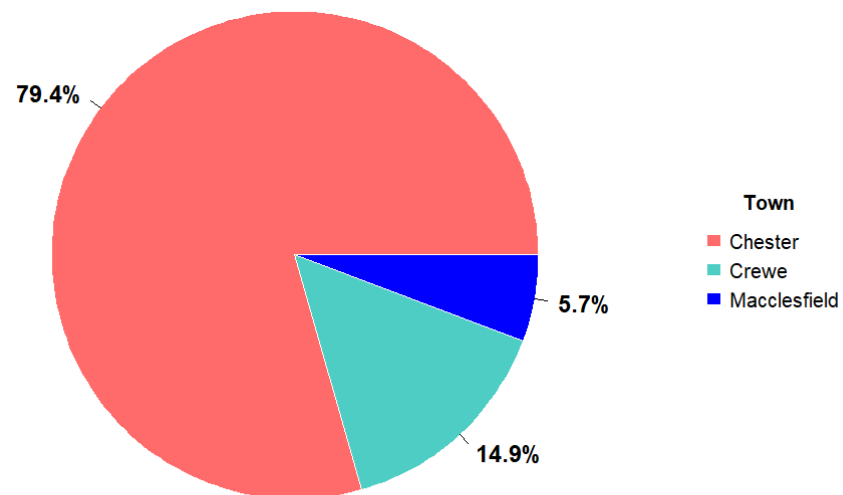


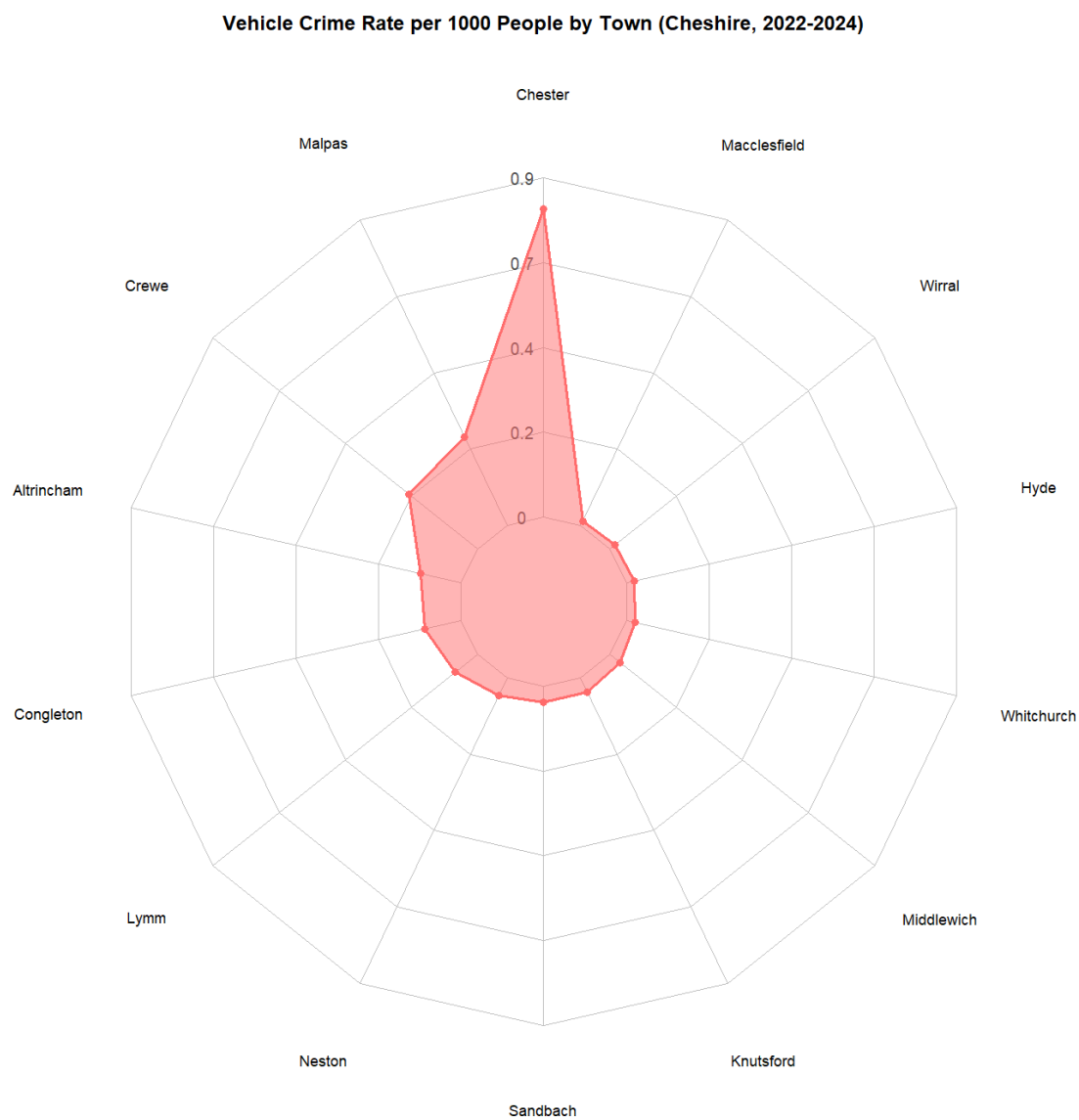
Figure 27: Code of Radar chart for Vehicle crime rate for Cheshire

```
# Create the radar chart
radarchart(
  radar_chart_data,
  axistype = 1,
  # Polygon customization
  pcol = "#FF6B6B",
  pfcpl = rgb(1, 0.42, 0.42, 0.5), # Semi-transparent fill
  plwd = 3,
  plty = 1,
  # Grid customization
  cglcol = "grey",
  cglty = 1,
  cglwd = 0.8,
  axislabcol = "grey30",
  # Label customization
  vlce = 0.9,
  caxislabels = round(seq(0, max_value, length.out = 5), 1),
  title = "Vehicle Crime Rate per 1000 People by Town (Cheshire, 2022-2024)"
)
dev.off()

# Print summary statistics
cat("\n=== Vehicle Crime Rate Summary for Cheshire Towns ===\n\n")
summary_table <- vehicle_rate_data %>%
  select(Town_formatted, vehicle_crime_count, population, vehicle_crime_rate) %>%
  arrange(desc(vehicle_crime_rate))

names(summary_table) <- c("Town", "Vehicle Crimes", "Population", "Rate per 1000")
print(summary_table)
```

Figure 28: Radar chart for Vehicle crime rate for Cheshire



(Radarchart Function - RDocumentation, n.d.)

Figure 29: Code for Line chart for Drug offense rates per 10,000 people for both counties

```
# LINE CHART: DRUG OFFENSE RATES (ALL YEARS)
# =====
cat("\n=== CREATING LINE CHART ===\n")
drug_rates <- crime_clean %>%
  filter(str_detect(crime_type, regex("Drug", ignore_case = TRUE))) %>%
  group_by(County, Year) %>%
  summarise(TotalCrimes = n(), .groups = "drop")

county_population <- district_populations %>%
  group_by(County) %>%
  summarise(TotalPopulation = sum(Pop2023, na.rm = TRUE), .groups = "drop")

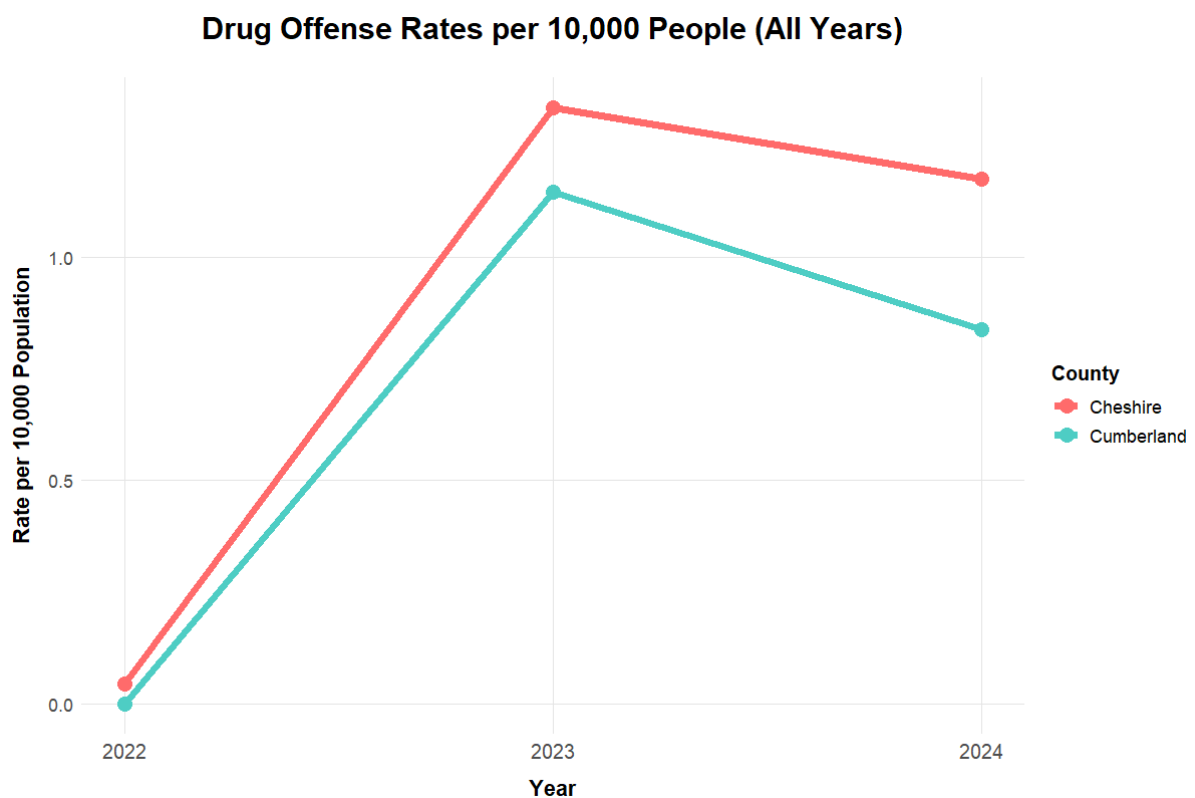
cat("County populations:\n")
print(county_population)

drug_rates <- drug_rates %>%
  left_join(county_population, by = "County") %>%
  filter(!is.na(TotalPopulation), TotalPopulation > 0) %>%
  mutate(Rate = (TotalCrimes / TotalPopulation) * 10000)

cat(paste0("\nDrug rate data rows: ", nrow(drug_rates), "\n"))

if(nrow(drug_rates) > 0) {
  p5 <- ggplot(drug_rates, aes(x = Year, y = Rate, color = County, group = County)) +
    geom_line(linewidth = 1.5) +
    geom_point(size = 3) +
    theme_minimal() +
    scale_color_manual(values = c("Cheshire" = "coral", "Cumberland" = "cyan3")) +
    labs(
      title = "Drug Offense Rates per 10,000 People (All Years)",
      x = "Year",
      y = "Rate per 10,000 Population",
      color = "County"
    )
}
```

Figure 30: Line chart for Drug offense rates per 10,000 people for both counties



Linear Model Visualization

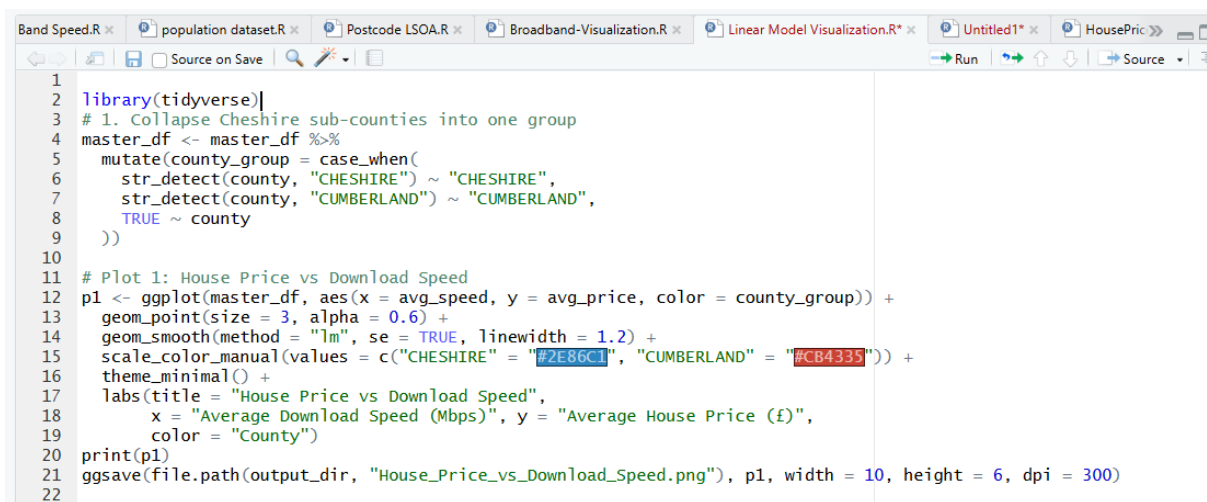
In this part, I use linear models to look at how house prices connect with important factors in the chosen areas. I use linear regression to find general trends and see how changes in things like time, population, and broadband speed affect property prices.

By making scatter plots and adding trend lines, I can better see patterns and compare different counties and towns. This method helps with clear, data-driven analysis and supports the final suggestions by showing the hidden connections in the information. ([Linear Models | Brilliant Math & Science Wiki](#), n.d.)

House Price vs Download Speed for both District (Cheshire and Cumberland)

R code

Figure 31: R Code of House Price vs Download Speed for both District (Cheshire and Cumberland)



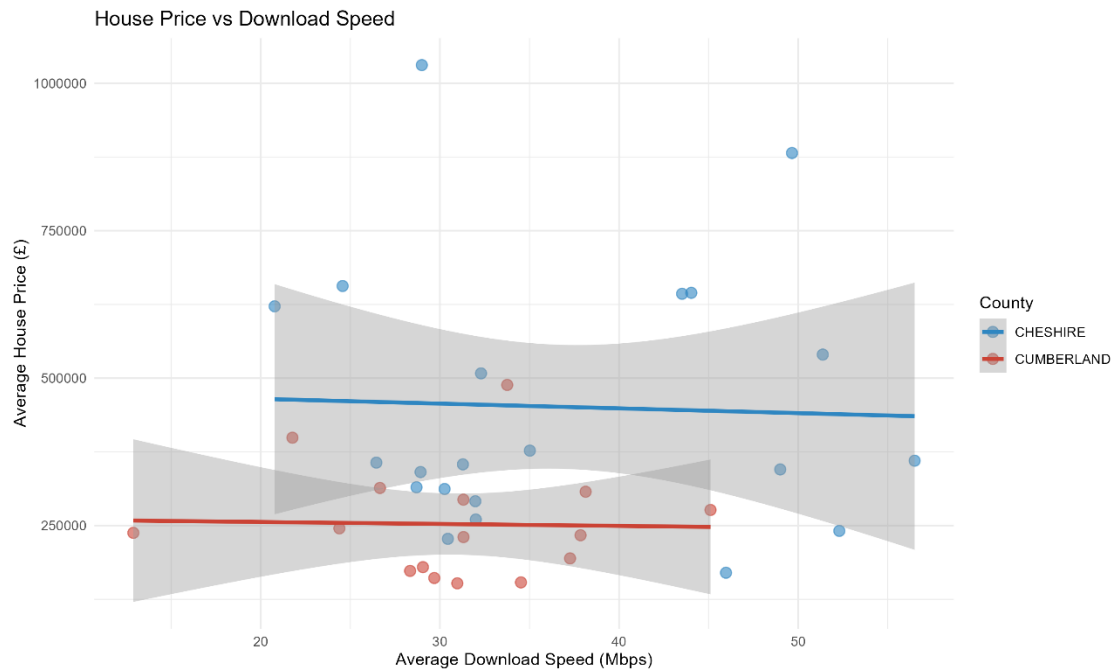
```

1  library(tidyverse)
2  # 1. Collapse Cheshire sub-counties into one group
3  master_df <- master_df %>%
4    mutate(county_group = case_when(
5      str_detect(county, "CHESHIRE") ~ "CHESHIRE",
6      str_detect(county, "CUMBERLAND") ~ "CUMBERLAND",
7      TRUE ~ county
8    ))
9
10
11 # Plot 1: House Price vs Download Speed
12 p1 <- ggplot(master_df, aes(x = avg_speed, y = avg_price, color = county_group)) +
13   geom_point(size = 3, alpha = 0.6) +
14   geom_smooth(method = "lm", se = TRUE, linewidth = 1.2) +
15   scale_color_manual(values = c("CHESHIRE" = "#2E86C1", "CUMBERLAND" = "#C84333")) +
16   theme_minimal() +
17   labs(title = "House Price vs Download Speed",
18        x = "Average Download Speed (Mbps)", y = "Average House Price (£)",
19        color = "County")
20 print(p1)
21 ggsave(file.path(output_dir, "House_Price_vs_Download_Speed.png"), p1, width = 10, height = 6, dpi = 300)
22

```

Graph

Figure 32: Graph of House Price vs Download Speed for both District (Cheshire and Cumberland)



Linear model summary report and correlation

Figure 33: Linear model summary report and correlation of House Price Vs. Download Speed

```

--- MODEL 1: PRICE vs SPEED ---
> print(summary(lm(avg_price ~ avg_speed, data = master_df)))

Call:
lm(formula = avg_price ~ avg_speed, data = master_df)

Residuals:
    Min       1Q   Median       3Q      Max
-225832 -127905  -63229   66130  679221

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  276243    125150   2.207  0.0339 *
avg_speed      2601      3519   0.739  0.4647
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 205200 on 35 degrees of freedom
Multiple R-squared:  0.01537,    Adjusted R-squared:  -0.01276
F-statistic: 0.5464 on 1 and 35 DF,  p-value: 0.4647

```

House prices vs drug offence rate 2023 both District (Cheshire and Cumberland) in a single diagram

R code

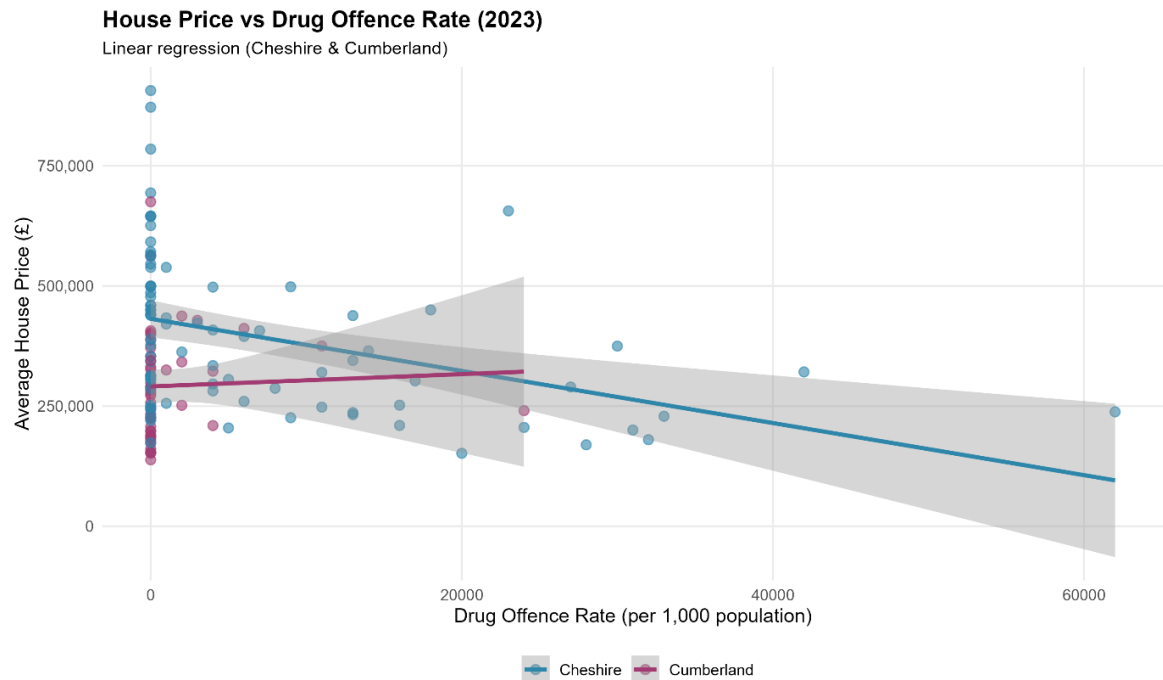
Figure 34: R Code of house prices vs drug offence rate 2023 both District (Cheshire and Cumberland) in a single diagram

```
# MODEL 2: House Price vs Drug Offence Rate (2023)
model2 <- lm(avg_price ~ drug_offence_rate * County, data = combined_data_clean)
print(summary(model2))

# Create visualization
p2 <- ggplot(combined_data_clean, aes(x = drug_offence_rate, y = avg_price, color = County)) +
  geom_point(alpha = 0.6, size = 2.5) +
  geom_smooth(method = "lm", se = TRUE, linewidth = 1.2) +
  scale_y_continuous(labels = comma, limits = c(NA, NA)) +
  scale_x_continuous(limits = c(NA, NA)) +
  scale_color_manual(values = c("CHESHIRE" = "#2E86AB", "CUMBERLAND" = "#A23B72"),
    labels = c("Cheshire", "Cumberland")) +
  labs(
    title = "House Price vs Drug Offence Rate (2023)",
    subtitle = "Linear regression (Cheshire & Cumberland)",
    x = "Drug Offence Rate (per 1,000 population)",
    y = "Average House Price (£)",
    color = ""
  ) +
  theme_minimal(base_size = 12) +
  theme(
    plot.title = element_text(face = "bold", size = 14),
    plot.subtitle = element_text(size = 11),
    legend.position = "bottom",
    panel.grid.minor = element_blank()
  )
```

Graph

Figure 35: Graph of house prices vs drug offence rate 2023 both District (Cheshire and Cumberland) in a single diagram



Linear model summary report and correlation

Figure 36: Linear model summary report and correlation of House Price Vs. Drug Offence Rate

```
>
> model2 <- lm(avg_price ~ drug_offence_rate * County, data = combined_data_clean)
> print(summary(model2))
```

Call:

```
lm(formula = avg_price ~ drug_offence_rate * County, data = combined_data_clean)
```

Residuals:

Min	1Q	Median	3Q	Max
-257990	-104266	-4770	77917	474875

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	4.317e+05	1.791e+04	24.108	< 2e-16 ***
drug_offence_rate	-5.420e+00	1.317e+00	-4.114	7.03e-05 ***
CountyCUMBERLAND	-1.408e+05	2.822e+04	-4.987	2.02e-06 ***
drug_offence_rate:CountyCUMBERLAND	6.703e+00	5.380e+00	1.246	0.215

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 139000 on 124 degrees of freedom
Multiple R-squared: 0.2064, Adjusted R-squared: 0.1872
F-statistic: 10.75 on 3 and 124 DF, p-value: 2.497e-06

Average download speed Vs. drug offence rate

R code

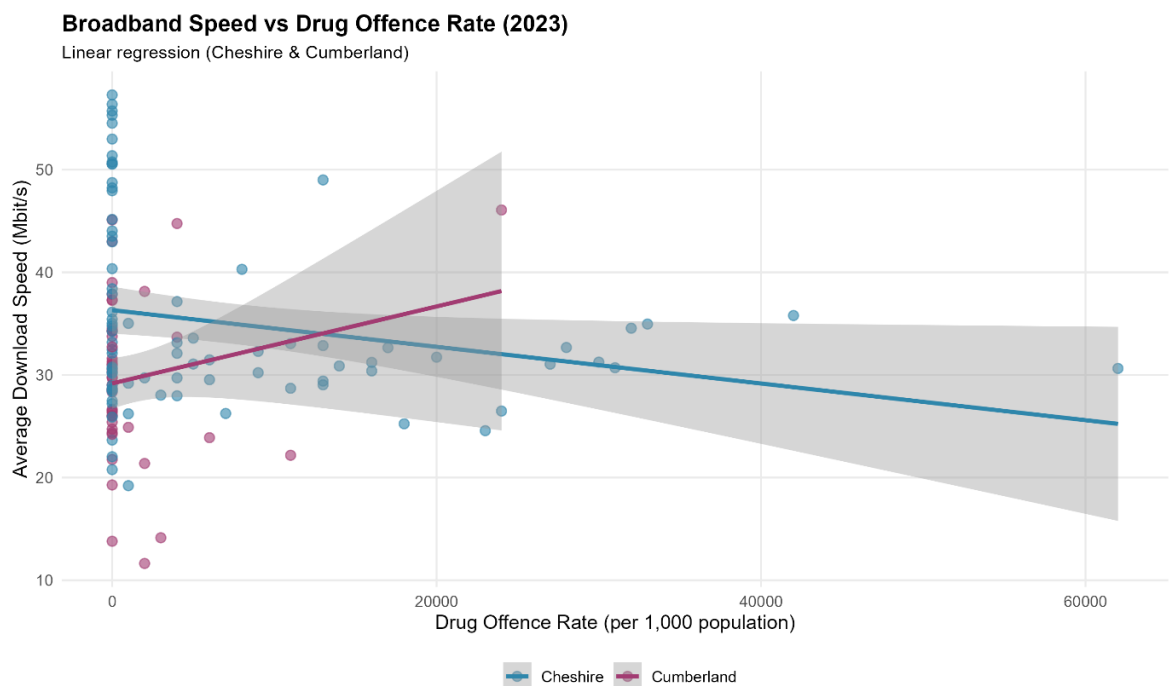
Figure 37: R Code of Average download speed Vs. drug offence rate

```
# MODEL 3: Download Speed vs Drug Offence Rate|
model3 <- lm(avg_download_speed ~ drug_offence_rate * County, data = combined_data_clean)
print(summary(model3))

# Create visualization
p3 <- ggplot(combined_data_clean, aes(x = drug_offence_rate, y = avg_download_speed, color = County)) +
  geom_point(alpha = 0.6, size = 2.5) +
  geom_smooth(method = "lm", se = TRUE, linewidth = 1.2) +
  scale_y_continuous(limits = c(NA, NA)) +
  scale_x_continuous(limits = c(NA, NA)) +
  scale_color_manual(values = c("CHESHIRE" = "#2E86AB", "CUMBERLAND" = "#A23872"),
    labels = c("Cheshire", "Cumberland")) +
  labs(
    title = "Broadband Speed vs Drug Offence Rate (2023)",
    subtitle = "Linear regression (Cheshire & Cumberland)",
    x = "Drug Offence Rate (per 1,000 population)",
    y = "Average Download Speed (Mbit/s)",
    color = ""
  ) +
  theme_minimal(base_size = 12) +
  theme(
    plot.title = element_text(face = "bold", size = 14),
    plot.subtitle = element_text(size = 11),
    legend.position = "bottom",
    panel.grid.minor = element_blank()
  )
```

Graph

Figure 38: Graph of Average download speed Vs. drug offence rate



Linear model summary report and correlation

Figure 39: Linear model summary report and correlation of Average download speed Vs. Drug Offence rate

```

> model3 <- lm(avg_download_speed ~ drug_offence_rate * County, data = combined_data_clean)
> print(summary(model3))

Call:
lm(formula = avg_download_speed ~ drug_offence_rate * County,
    data = combined_data_clean)

Residuals:
    Min       1Q   Median       3Q      Max
-18.287  -5.663  -1.212   5.113  20.970

Coefficients:
                Estimate Std. Error t value Pr(>|t|)
(Intercept)      3.631e+01  1.102e+00  32.949  < 2e-16 ***
drug_offence_rate -1.787e-04  8.109e-05  -2.204   0.0294 *
CountyCUMBERLAND  -7.142e+00  1.737e+00  -4.112  7.09e-05 ***
drug_offence_rate:CountyCUMBERLAND  5.537e-04  3.311e-04   1.672   0.0970 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 8.553 on 124 degrees of freedom
Multiple R-squared:  0.1253,    Adjusted R-squared:  0.1041
F-statistic:  5.92 on 3 and 124 DF,  p-value: 0.0008289

```

Recommendation System

Overview

The recommendation system created in this project is meant to give a clear, data-based list of the best towns for investing in property in Cheshire and Cumberland. The system is built using R with a well-organized and open method for analysis. I start by loading the cleaned data and using the tidy verse group of tools to handle and combine the data efficiently.

Functions like `read_csv()` are used to bring in the prepared data sets, and `left join()` is used to put together information about housing, broadband, crime, and population at the same geographical level. Any missing data that's left is removed to make sure the final results are consistent and trustworthy. This organized way of working lets different factors be combined into one overall score that helps make fair and based-on-evidence choices.

Result

The results shown in the RStudio console list the top ten towns according to their recommendation scores. These results show how well each town performs in important areas like house price, broadband speed, and crime rate. The ranking shows towns that do a good job of balancing cost, internet quality, and safety. Overall, the results show big differences between towns, with the top ones usually having lower house prices, better broadband, and less crime. These findings give a clear and easy-to-understand way to find the best towns for investing.

Figure 40: Top 10 House Score

TOP 10 TOWNS BY HOUSE SCORE (Affordability)		
#	Town <chr>	HouseScore <dbl>
1	MOOR ROW	10.00
2	MILLOM	9.99
3	MARYPORT	9.99
4	EGREMONT	9.99
5	CLEATOR	9.98
6	WIDNES	9.98
7	WHITEHAVEN	9.98
8	WORKINGTON	9.97
9	CLEATOR MOOR	9.97
10	FRIZINGTON	9.97

Figure 41: Top 10 Crime Score

```
=====
TOP 10 TOWNS BY CRIME SCORE (Safety)
=====
```

#	Town <chr>	CrimeScore <dbl>

1	Amber Valley 001B	10.00
2	Arun 013C	10.00
3	Babergh 011C	10.00
4	Barnsley 025C	10.00
5	Basildon 015C	10.00
6	Basingstoke and Deane 011B	10.00
7	Bassetlaw 008F	10.00
8	Birmingham 050E	10.00
9	Birmingham 094C	10.00
10	Blackburn with Darwen 006E	10.00

Figure 42: Top 10 Broadband Speed Score

```
=====
TOP 10 TOWNS BY BROADBAND SCORE
=====
```

#	Town <chr>	BroadbandScore <dbl>

1	CHEADLE	10.00
2	MANCHESTER	8.39
3	WIRRAL	7.73
4	STOCKPORT	7.11
5	ELLESMERE PORT	6.82
6	ALDERLEY EDGE	6.44
7	ALTRINCHAM	6.25
8	RUNCORN	6.10
9	WILMSLOW	6.03
10	MACCLESFIELD	5.88

Reflection

Working on this project has been a really valuable experience. I got to gather, clean, analyse, and interpret data, which helped me learn a lot. One of the main challenges was dealing with data from different sources, since each had its own issues like missing information and different formats. Cleaning the data needed a lot of attention to make sure everything was consistent and could be put together properly.

By looking at the data closely, I noticed patterns in things like broadband speeds, crime rates, and house prices. Making charts and visuals helped me spot trends, strange things, and connections that weren't obvious in the raw numbers. Using statistical models let me measure how different factors affect each other, which really deepened my understanding. To show the results more clearly, I made a scoring system for towns based on house price, broadband, and crime. Each category was scored from 0 to 10, with 10 being the best. I've attached the top 10 towns based on this combined score, so you can see which areas do best in these three areas.

This project helped me get better at using tools like R, making data visualizations, and building statistical models. It also made me think more critically and solve problems more effectively. Overall, it showed me how important clean data and good analysis are, and how important it is to explain results clearly. It gave me confidence to use data to solve real-world problems in the future. If you'd like, I can also make a simpler, more personal version of this that's great for submitting with your top 10 towns attached. It'll feel more like your own thoughts rather than a formal report.

House Prices

The CleanedHousePrices.csv dataset was used to create these rankings. A score of 10 means the area offers the best value for money compared to the average property cost in the town.

Figure 43: House Price Recommendation

	Town	HouseScore	avg_price	Recommendation
1	MOOR ROW	10.000000	126741.9	Affordable
2	MILLOM	9.989943	147130.6	Affordable
3	MARYPORT	9.987510	152063.0	Affordable
4	EGREMONT	9.986737	153630.1	Affordable
5	CLEATOR	9.983067	161071.3	Affordable
6	WIDNES	9.978662	170000.0	Affordable
7	WHITEHAVEN	9.977125	173117.9	Affordable
8	WORKINGTON	9.973968	179517.9	Affordable
9	CLEATOR MOOR	9.971880	183750.3	Affordable
10	FRIZINGTON	9.966644	194366.4	Affordable

Broadband Speeds

This table looks at digital infrastructure, helping people find places with strong internet connections, such as Cheadle and Manchester.

Figure 44: Broadband Speed score and Recommendation

	Town	BroadbandScore	avg_download	avg_upload	Recommendation
1	CHEADLE	10.000000	67.01509	6.376887	Good
2	MANCHESTER	8.387929	59.56059	5.982353	Good
3	WIRRAL	7.726522	56.50213	6.198652	Good
4	STOCKPORT	7.110671	53.65432	6.627656	Good
5	ELLESMERE PORT	6.817120	52.29688	6.243947	Good
6	ALDERLEY EDGE	6.436873	50.53855	7.878916	Good
7	ALTRINCHAM	6.245784	49.65493	8.056743	Moderate
8	RUNCORN	6.103893	48.99880	6.555422	Moderate
9	WILMSLOW	6.032839	48.67023	6.960177	Moderate
10	MACCLESFIELD	5.877954	47.95401	6.146381	Moderate

Crimes

Safety is measured on a scale from 0 to 10, with 10 being the safest.

These areas have very low crime rates, which gives them a "Safe" rating.

Figure 45: Crime (Safe Town) Recommendation

	Town	CrimeScore	total_crimes	Recommendation
1	Amber Valley 001B	10	1	Safe
2	Arun 013C	10	1	Safe
3	Babergh 011C	10	1	Safe
4	Barnsley 025C	10	1	Safe
5	Basildon 015C	10	1	Safe
6	Basingstoke and Deane 011B	10	1	Safe
7	Bassetlaw 008F	10	1	Safe
8	Birmingham 050E	10	1	Safe
9	Birmingham 094C	10	1	Safe
10	Blackburn with Darwen 006E	10	1	Safe

Overall Score

The overall recommendation for each town in the dataset is based on an overall_score. This score is made by combining and adjusting four important factors: average broadband speed, housing affordability, crime rates, and broadband quality. The final overall_score is the average of these four adjusted scores. Using this detailed analysis, the table shows how towns are ranked. The top towns, Cheadle, Wirral, and Stockport, have an “Excellent” rating. These towns have good digital connections, low crime, and affordable housing. On the other hand, towns like Carlisle and Warrington have lower scores and are rated “Needs Improvement” because they don’t perform well in one or more areas.

Figure 46: Overall Score

Town	avg_download_speed	BroadbandScore	HouseScore	avg_house_price	total_crimes	CrimeScore	overall_score	Recommendation
1 CHEADLE	67.01509	10.000000	9.870145	390000.0	0	9.996145	9.955430	Excellent
2 WIRRAL	56.50213	7.7265225	9.884943	360000.0	17	9.997430	9.202965	Excellent
3 STOCKPORT	53.65432	7.1106713	9.835019	461213.0	15	9.998001	8.981230	Excellent
4 ELLESMERE PORT	52.29688	6.8171200	9.943652	240977.0	0	9.996145	8.918972	Excellent
5 ALDERLEY EDGE	50.53855	6.4368730	9.675438	784734.8	0	9.996145	8.702819	Excellent
6 RUNCORN	48.99880	6.1038931	9.892242	345201.5	0	9.996145	8.664094	Excellent
7 ALTRINCHAM	49.65493	6.2457845	9.627575	881767.9	0	9.996145	8.623168	Excellent
8 WILMSLOW	48.67023	6.0328389	9.812265	507342.9	0	9.996145	8.613749	Excellent
9 MACCLESFIELD	47.95401	5.8779539	9.874025	382134.6	0	9.996145	8.582708	Excellent
10 WIDNES	45.97059	5.4490295	9.978662	170000.0	0	9.996145	8.474612	Excellent
11 BECKERMET	45.10476	5.2617906	9.926142	276476.2	0	9.996145	8.394693	Excellent
12 LYMM	44.03410	5.0302541	9.744516	644691.1	0	9.996145	8.256972	Excellent
13 ST BEES	43.00931	4.8086400	9.936142	256203.9	0	9.996145	8.246975	Excellent
14 KNUTSFORD	43.51932	4.9189308	9.744339	645050.6	0	9.996145	8.219805	Excellent
15 HYDE	38.39745	3.8113038	9.951533	225000.0	0	9.996145	7.919661	Consider Improvement
16 BROUGHTON-IN-FURNESS	39.00107	3.9418393	9.785469	561666.7	0	9.996145	7.907818	Consider Improvement
17 SEASCALE	37.85133	3.6932023	9.947298	233586.9	0	9.996145	7.878882	Consider Improvement
18 MOOR ROW	37.31623	3.5774847	10.000000	126741.9	0	9.996145	7.857877	Consider Improvement
19 FRIZINGTON	37.26527	3.5664641	9.966644	194366.4	0	9.996145	7.843084	Consider Improvement
20 EGREMONT	34.52350	2.9735450	9.986737	153630.1	0	9.996145	7.652142	Consider Improvement
21 NESTON	35.02052	3.0810268	9.876518	377080.0	0	9.996145	7.651230	Consider Improvement
22 CLEATOR MOOR	34.28497	2.9219615	9.971880	183750.3	0	9.996145	7.629995	Consider Improvement
23 STOKE-ON-TRENT	33.64387	2.7833199	9.915469	298114.8	0	9.996145	7.564978	Consider Improvement
24 KESWICK	33.75678	2.8077384	9.821570	488478.4	0	9.996145	7.541818	Consider Improvement
25 CREWE	32.22483	2.4764471	9.933508	261542.8	0	9.996145	7.468700	Consider Improvement
26 COCKERMOUTH	31.30966	2.2785361	9.917515	293965.1	0	9.996145	7.397399	Consider Improvement
27 MARYPORT	30.96986	2.2050528	9.987510	152063.0	0	9.996145	7.396236	Consider Improvement
28 NORTHWICH	30.73285	2.1537997	9.917920	293144.4	0	9.996145	7.355955	Consider Improvement
29 CHESTER	30.62929	2.1314044	9.887984	353833.9	0	9.996145	7.338511	Consider Improvement
30 SANDRACH	30.26017	2.0515801	9.908502	312056.0	0	9.996145	7.318772	Consider Improvement

Showing 1 to 30 of 54 entries, 9 total columns

30	SANDBACH	30.26017	2.0515801	9.908592	312056.0	0	9.996145	7.318772	Consider Improvement
31	CLEATOR	29.68571	1.9273512	9.983067	161071.3	0	9.996145	7.302188	Consider Improvement
32	FRODSHAM	29.61680	1.9124482	9.894446	340734.0	0	9.996145	7.267680	Consider Improvement
33	WORKINGTON	29.05057	1.7899988	9.973968	179517.9	0	9.996145	7.253370	Consider Improvement
34	MIDDLEWICH	28.99070	1.7770507	9.893404	342846.2	0	9.996145	7.222200	Consider Improvement
35	MILLOM	28.47586	1.6657144	9.989943	147130.6	0	9.996145	7.217267	Consider Improvement
36	CONGLETON	28.69360	1.7128027	9.907157	314965.0	0	9.996145	7.205368	Consider Improvement
37	WHITEHAVEN	28.33456	1.6351589	9.977125	173117.9	0	9.996145	7.202809	Consider Improvement
38	PENRITH	28.60114	1.6928082	9.907756	313750.0	0	9.996145	7.198903	Consider Improvement
39	WINSFORD	28.33945	1.6362165	9.948729	230685.8	0	9.996145	7.193697	Consider Improvement
40	BRAMPTON	28.30591	1.6289631	9.910887	307403.5	0	9.996145	7.178665	Consider Improvement
41	MALPAS	28.42398	1.6544956	9.840189	450730.5	0	9.996145	7.163610	Consider Improvement
42	HIGH PEAK	28.49730	1.6703506	9.743941	645857.1	8	10.000000	7.138097	Consider Improvement
43	WHITCHURCH	27.48636	1.4517304	9.784997	562622.9	0	9.996145	7.077624	Consider Improvement
44	NANTWICH	26.45088	1.2278031	9.886587	356666.5	0	9.996145	7.036845	Consider Improvement
45	NEWCASTLETON	26.33643	1.2030529	9.892443	344795.0	0	9.996145	7.030547	Consider Improvement
46	TARPORLEY	26.53392	1.2457604	9.797929	536406.1	0	9.996145	7.013278	Consider Improvement
47	AMBLESIDE	26.50775	1.2401024	9.729566	675000.0	0	9.996145	6.988604	Consider Improvement
48	WIGTON	24.68876	0.8467374	9.945296	237643.8	0	9.996145	6.929393	Consider Improvement
49	RAVENGLASS	24.38667	0.7814080	9.941522	245296.4	0	9.996145	6.906358	Consider Improvement
50	HOLMROOK	21.76667	0.2148211	9.865707	398997.7	0	9.996145	6.692224	Consider Improvement
51	TIVERTON	20.77330	0.0000000	9.919471	290000.0	0	9.996145	6.638539	Consider Improvement
52	MANCHESTER	59.56059	8.3879290	0.000000	20400000.0	26	9.994860	6.127596	Consider Improvement
53	CARLISLE	31.49934	2.3195564	9.948812	230517.6	24850	2.906137	5.058168	Consider Improvement
54	WARRINGTON	37.97064	3.7190033	9.740189	653464.3	35027	0.000000	4.486397	Needs Improvement

Showing 25 to 54 of 54 entries, 9 total columns

Legal and Ethical Issues

The creation of this recommendation system needs attention to important legal and ethical aspects to ensure that data is used in a responsible and clear way. One big legal issue is protecting people's privacy. The system uses data about house prices, broadband speed, and crime rates, which could be risky if it looks at personal information. To keep things safe, all the data was grouped together and made anonymous at the town or district level. This way, no personal details are shared, and it follows rules for protecting data.

From an ethical point of view, the way the scoring and recommendations are made has some possible problems. The system uses certain factors like broadband, house prices, and crime, and gives each the same weight when scoring on a scale from 0 to 10, with 10 being the best. This makes it easier to compare, but it might not be fair. Giving equal weight to all these factors could make some areas look better than others, even if they have different needs. It's important to remember that not all areas are the same, and equal weighting might not show the full picture of real life. ([Lee, 2024](#))

Another ethical issue is how people might misunderstand the results. The overall score is just a number that represents many different social and economic factors. So labels like “Highly Recommended” should not be taken too seriously. A lower score in one area, like crime or broadband, doesn't mean a whole town or neighbourhood is bad. It's important to include warnings so people don't draw wrong conclusions. ([Dorn et al., n.d.](#))

Lastly, being open and clear is key to making people trust the system. All the data, ideas, and steps used to get the final scores have been written down clearly. This openness helps users and others understand and check how the system works, making sure it's honest and can be repeated by others.

Conclusion

This project used the whole data science process to help international investors decide between Cheshire and Cumberland for property investment. I got data from UK government sources and combined information on house prices, broadband speed, crime rates, and access to local services. The data was cleaned to fix any mistakes, make sure all the information was in the same format, and keep the data organized for better storage. I made sure the data met a standard level of quality called third normal form (3NF). I also dealt with any strange or missing data points and only included towns that matched the right population size and administrative area. I used R to look at the data closely, making charts like histograms, boxplots, and correlation tables to find out what was connected between different factors. Each town was given a score from 0 to 10 based on four main areas: how much it costs to buy property, how well connected it is, how safe it is, and what services are available. I showed these scores with bar charts and ranked lists. The system suggested top areas like Cheadle, Wirral, and Stockport, which had good broadband, low crime, and good property values. While Cumberland areas were usually safer and cheaper, Cheshire areas had better internet and more developed infrastructure. This approach gives a clear and repeatable way to check how good a place is to live and helps investors make better decisions about where to buy property.

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Appendix**GitHub Link**

[Neehangma/Data-Science_Neehangma](#)

Google Drive link

https://drive.google.com/drive/folders/185uyIXN0BqtKQ4uQ6J6gMU92_0qGemR?usp=drive_link